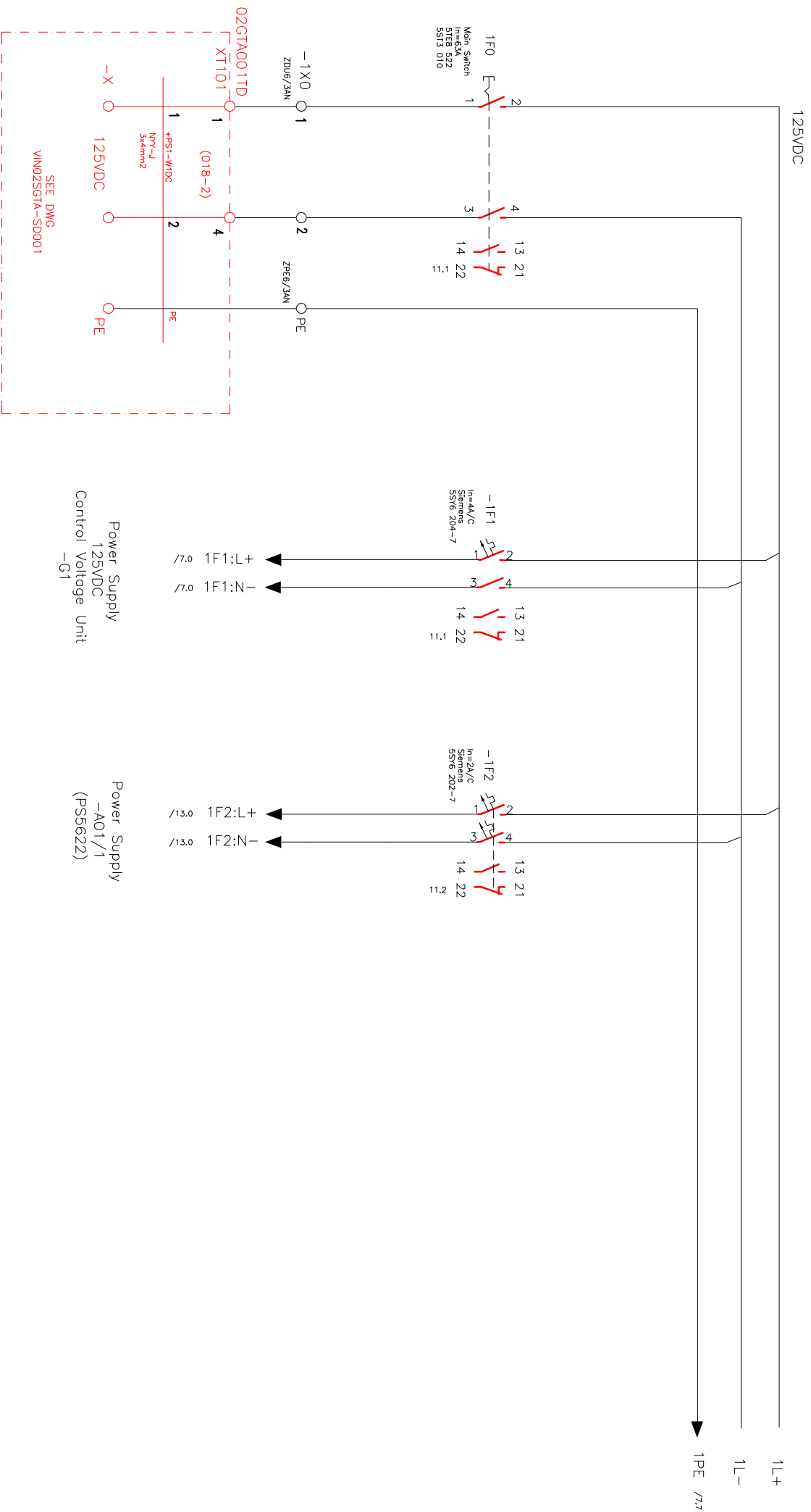
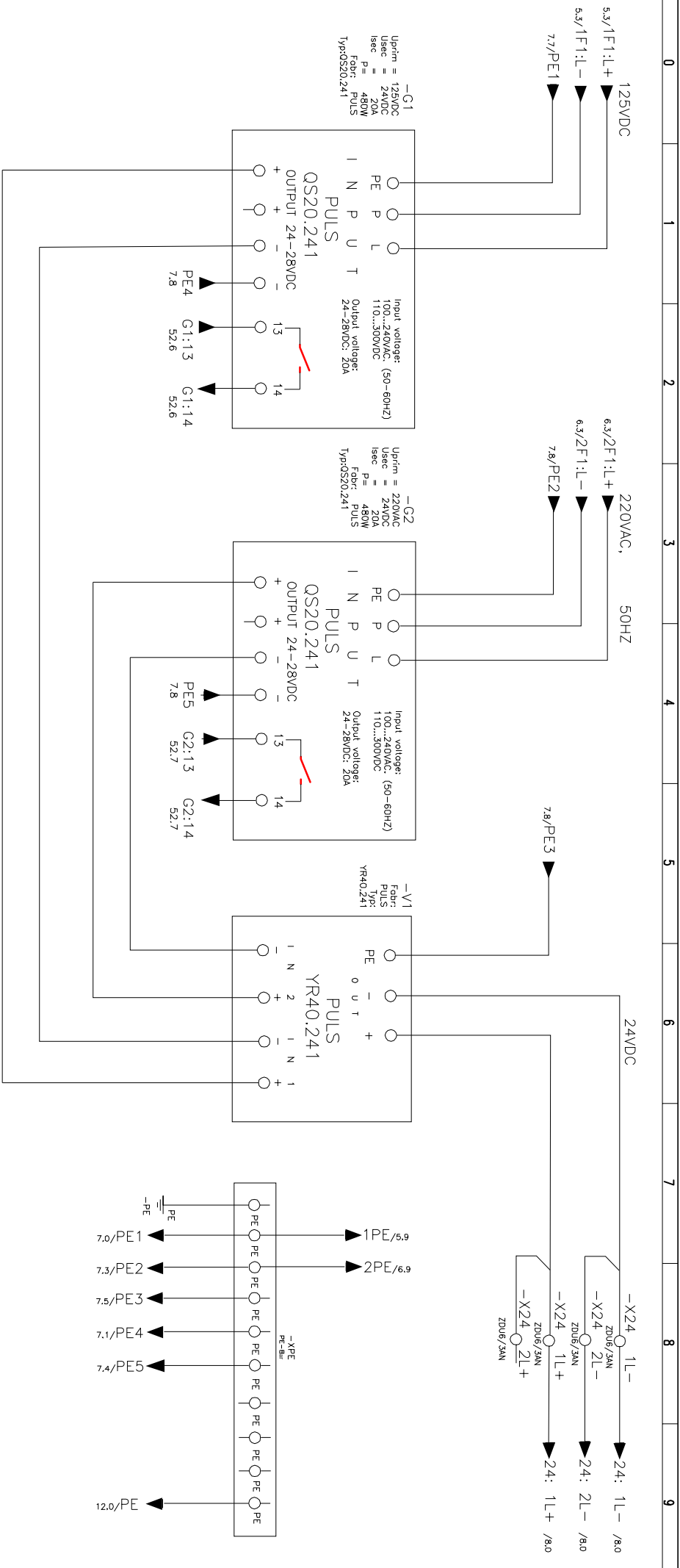




[illegible]



This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Patenzugriff: 76 1141 Vienna Austria-11, project: not 50 16016 Diese Zeichnung ist EIGENTUM der ANDRITZ HYDRO GmbH Patenzugriff: 76 1141 Wien, ÖSTERREICH-Schultz, 50 16016 beachten		Gez./Draw 20.06.2016 Auer		Kunde/Cient: VINH SON	
Date Zeichnung 20.06.2016 Fuchs		Gez./Check 20.06.2016 Fuchs		Anlage/Plant: VINH SON	
Anderrung / Modification Datum/Date Beort./Des. Mossstab/Sole		Gez./Apr. 20.06.2016 Fuchs		Projec - No : Lost Edit: 11.07.2012	
		TURBINE CONTROL CUBICLE Circuit Diagramm Control Voltage Units 110VDC/24VDC, 110VDC/24VDC.			
Zeich./Draw No. WIN12-60W-0021		Index Bl./Sh. 7 von/of 69		= VINH SON +ME100H100	
				A3	

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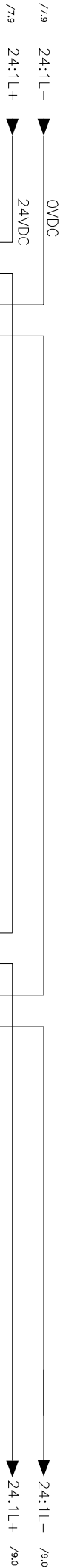
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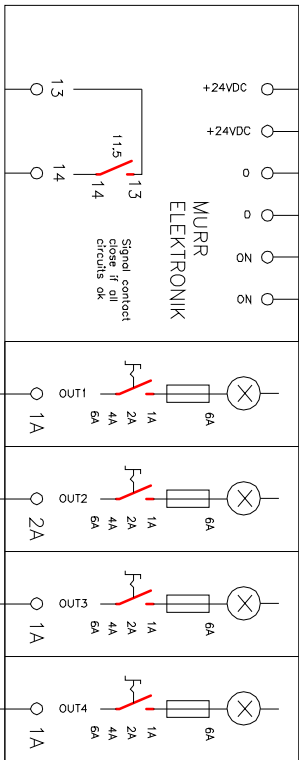
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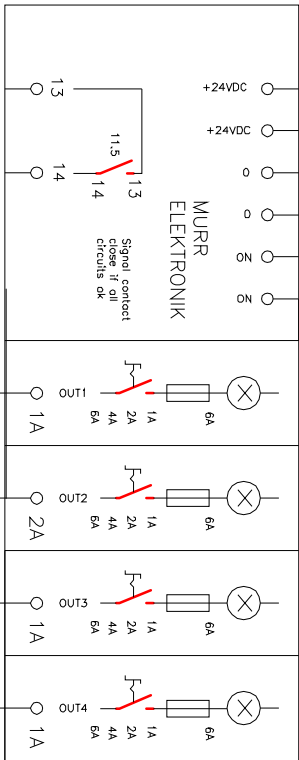
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—FO1
Load Control
4-channel
IN: 24VDC
OUT: 24VDC
1-2-4-6A

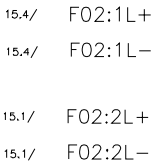
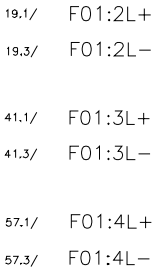


—FO2
Load Control
4-channel
IN: 24VDC
OUT: 24VDC
1-2-4-6A



—X24
ZDU2.5/3AN

—X24
ZDU2.5/3AN



Power Supply
F01:1L+ SPARE
F01:2L+ Control Unit Rail 0
F01:3L+ Control Unit Rail 1
F01:4L+ Control Unit Rail 2

Power Supply
F02:1L+ Touch Panel
F02:2L+ Ethernet Switch
F02:3L+ SPARE
F02:4L+ SPARE

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				Gepr./Check	20.06.2016	Fuchs	Anlage/Plant:					24VDC Control Voltage Distribution		Zech./Draw.No.	+MEY008100	
				Ges./Appr.	20.06.2016	Fuchs	Projec - No :					Lead Circuit Controls		Index	B1./ShL 8	
Änderung / Modification				Datum/Date			Lost Edit: 11.05.2012							von/of 69	A3	

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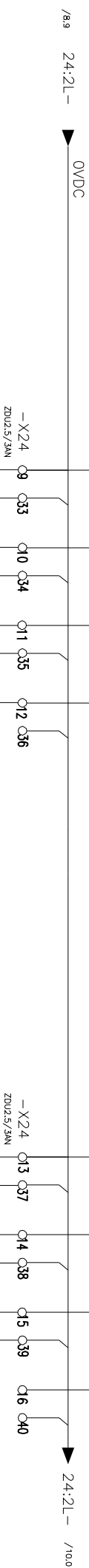
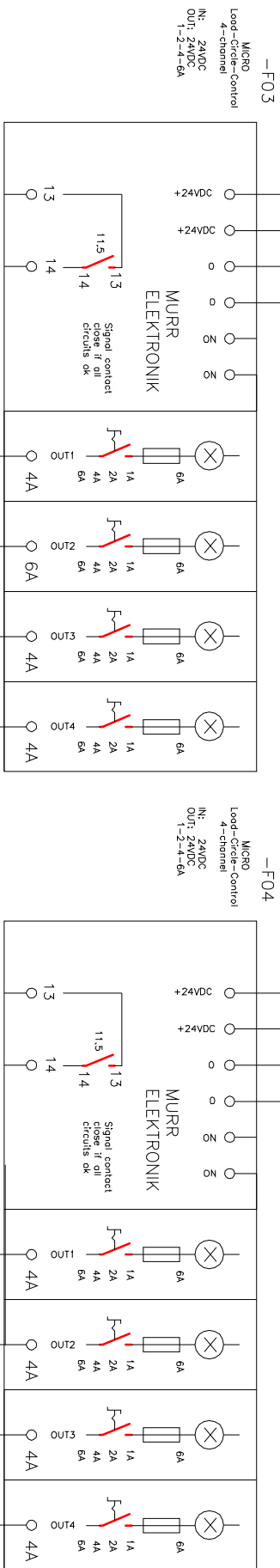
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Power Supply
F03:1L+ Amplifier signal Needle1
F03:2L+ Amplifier signal Deflector1
F03:3L+ Amplifier signal Needle4
F03:4L+ OXDI:65

Power Supply
F04:1L+ Amplifier signal Needle1
F04:2L+ SPARE
F04:3L+ Amplifier signal Needle3
F04:4L+ SPARE

This drawing is the property of ANDRITZ HYDRO GmbH Patentreferiert/26 1141 Vanno Austria-rl, project, note ISO 16016 Diese Zeichnung ist GEISTIGES EIGENTUM der ANDRITZ HYDRO GmbH Patentreferiert/26 1141 Wirt. OSTERREICH-Schultz, ISO 16016 beachten				Gez./Draw	20.06.2016	Auer	Kunde/Client:	ANDRITZ HYDRO Hydro			
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				Ges./Appr.	20.06.2016	Fuchs	Projec - No :				
				Bearb./Des.			Lost Edit:	11.05.2012			
Änderung / Modification				Datum/Date			Mossstab/Scale	-----			
								TURBINE CONTROL CUBICLE			
								Circuit Diagram			
								24VDC Control Voltage Distribution			
								Load Circuit Controls			
								Int.No			
								= VINH SON			
								+METROCH100			
								Index			
								Bl./Sht. 9			
								von/of 69			
								A3			

9.9/ 24:1L- 24VDC

9.9/ 24:1L+ 24VDC

-F05

MICO

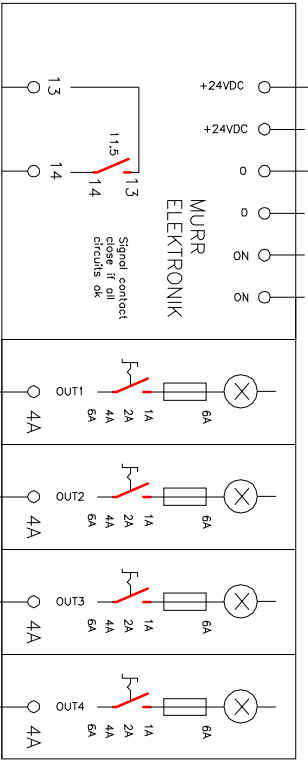
Load-Control

4-channel

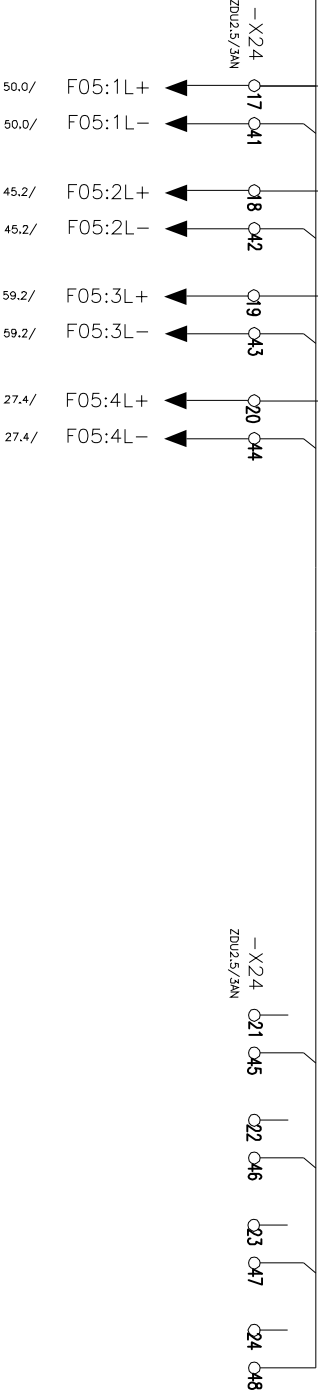
IN: 24VDC

OUT: 24VDC

1-2-4-6A



/9.9 24:2L- OVDC



Power Supply

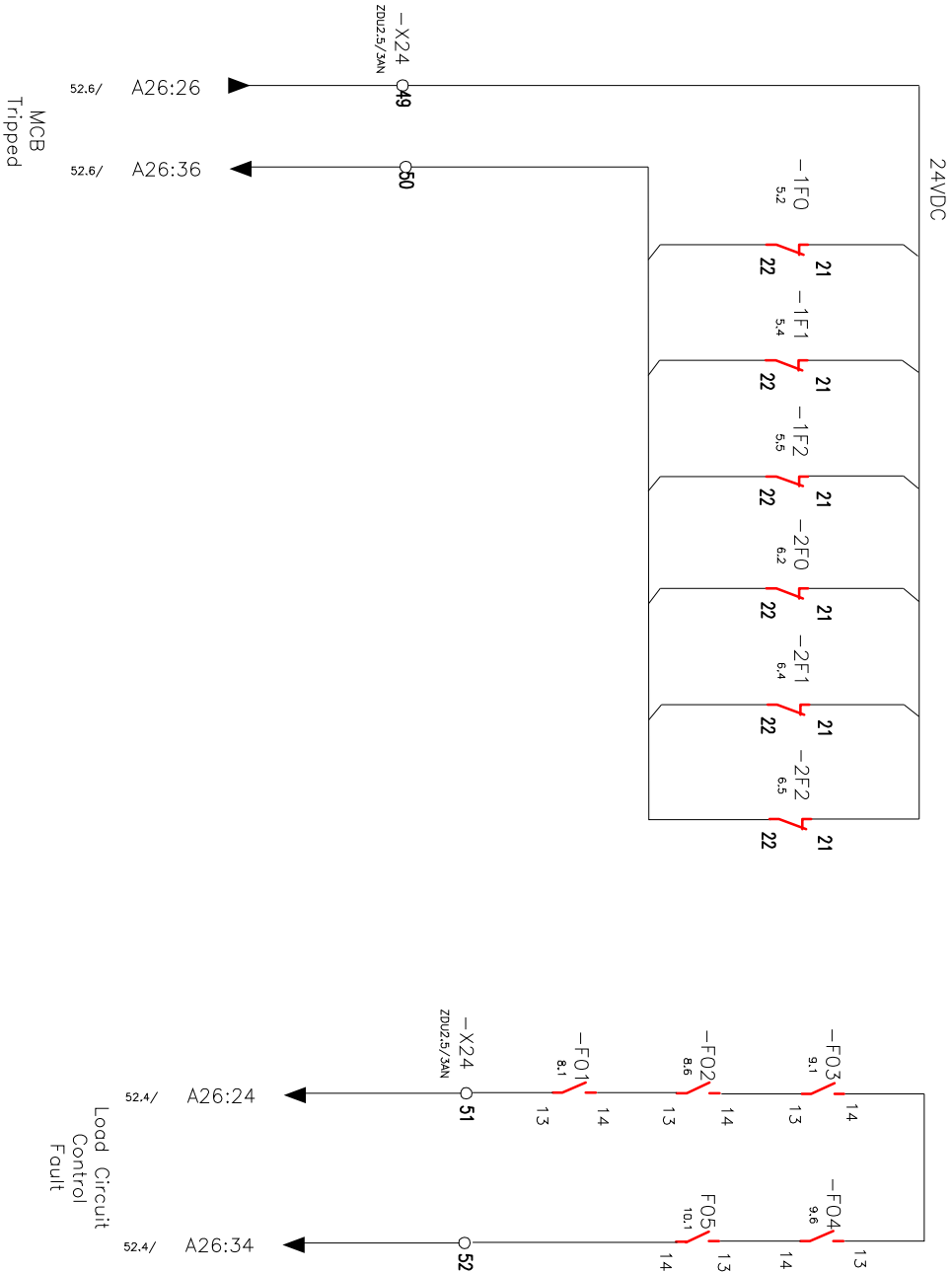
F05:1L+ Signal Insulator

F05:2L+ Isolating Amplifier Rai #1

F05:3L+ Isolating Amplifier Rai #2

F05:4L+ Sineax P530

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Gepr./Check 20.06.2016				Fuchs		Anlage/Plant: VINH SON						Circuit Diagram		Zech-/Draw.No.		+MET008100			
Ges./Appr. 20.06.2016				Fuchs		Projec - No : -----						Lost Edit: 20.07.2016		Index		Bl./Sht. 10			
Änderung / Modification				Datum/Date		Beorb./Des.		Mossstab/Sole		Projec - No : -----		Lost Edit: 20.07.2016		VIN12-CON-0021		von/of 69		A3	

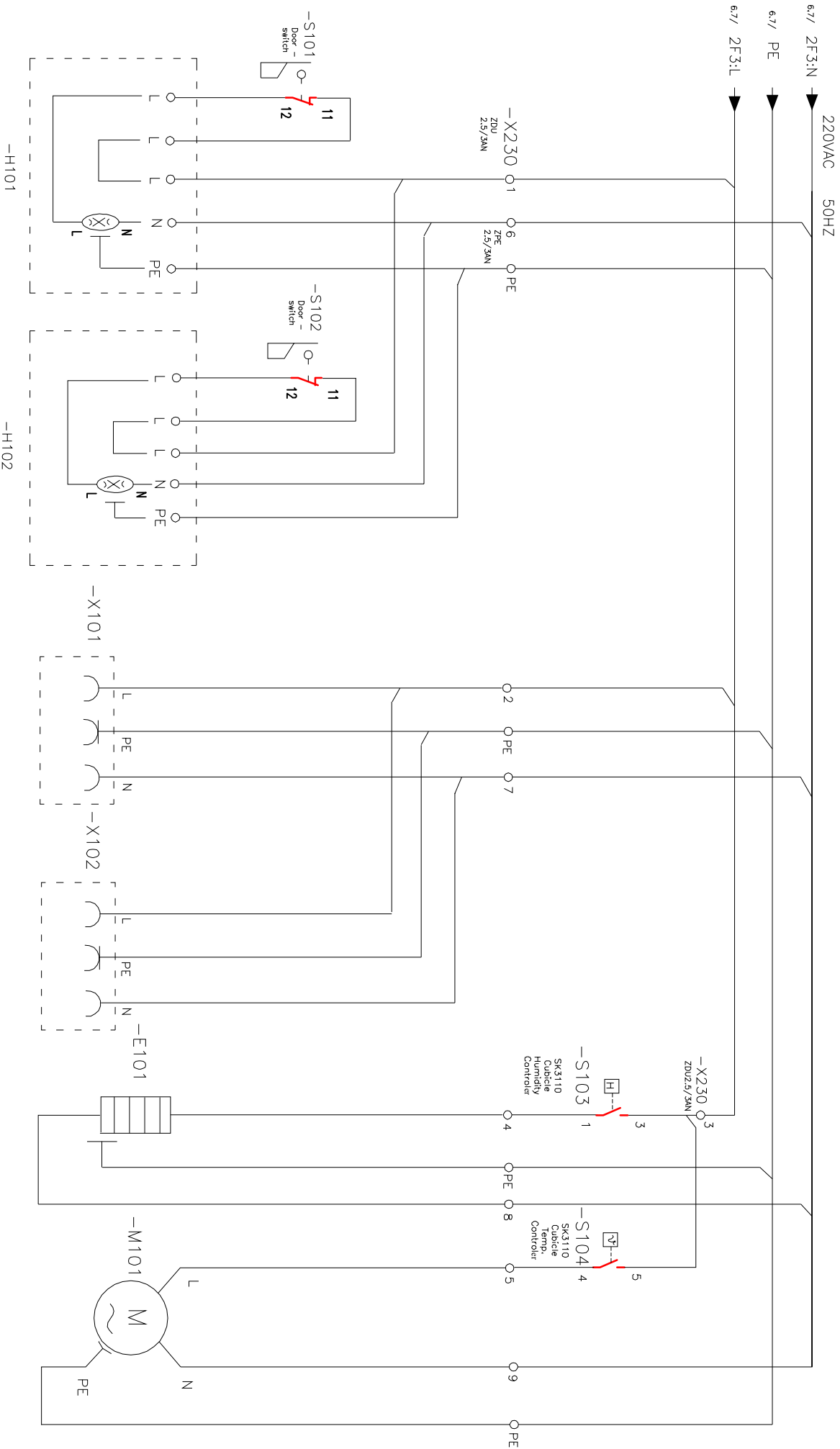


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				Ges./Appr.	20.06.2016	Fuchs	Projec - No :
				Masstab/Sole	-----		Lost Edit: 20.07.2016
Änderung / Modification				Datum/Date	Beorb./Des.		

ANDRITZ
HYDRO
Hydro

TURBINE CONTROL CUBICLE
Circuit Diagram
Fuse Failure. Load Circuit Controls

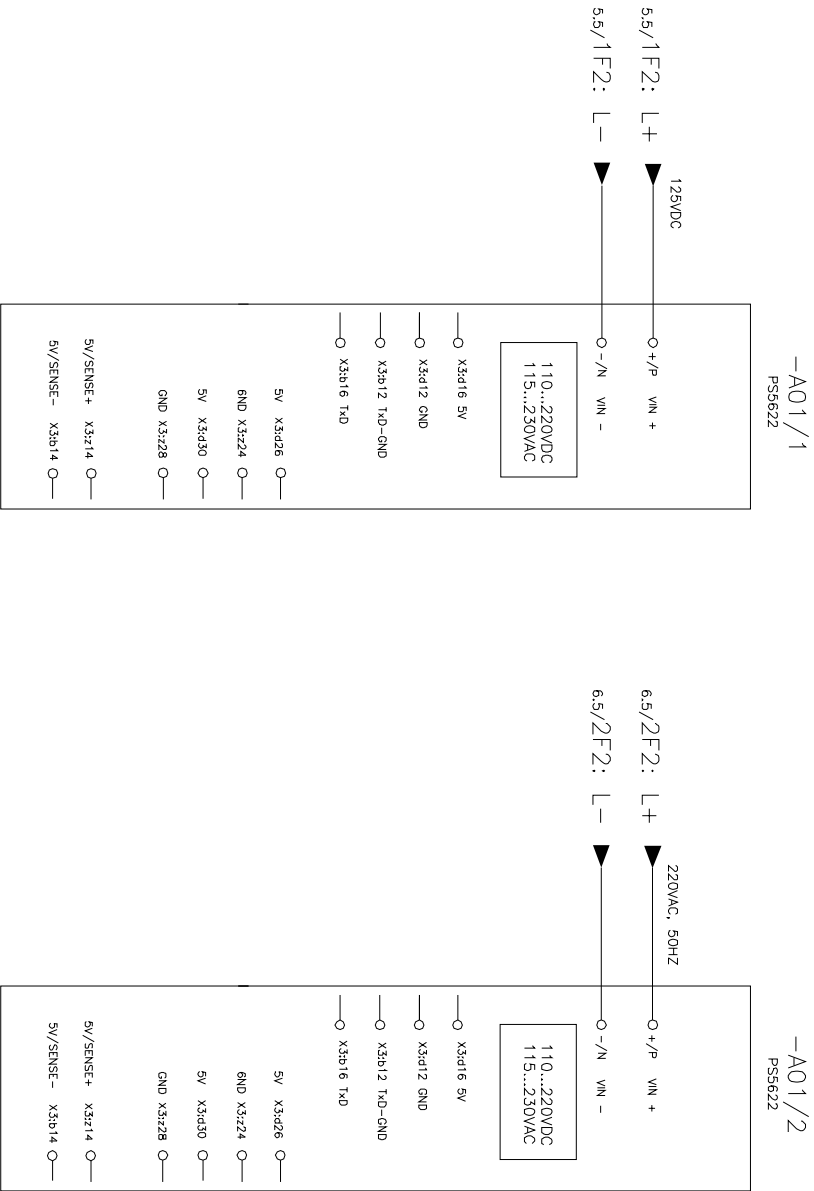
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Zeich./Draw.No.		+MEY10GH100	
VN12-CON-0021		Index	Bl./Sht. 11
		von/of 69	
		A3	



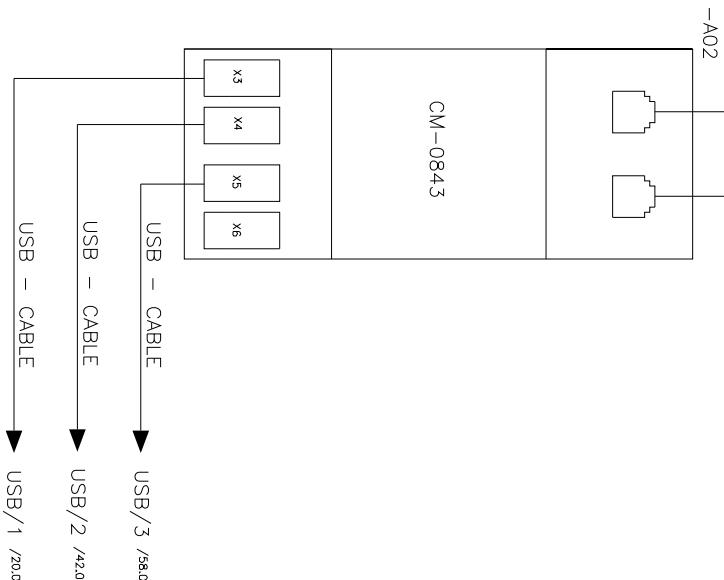
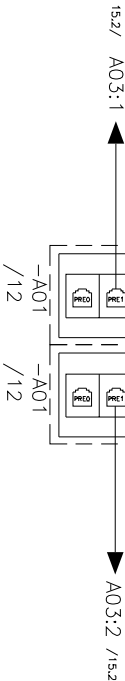
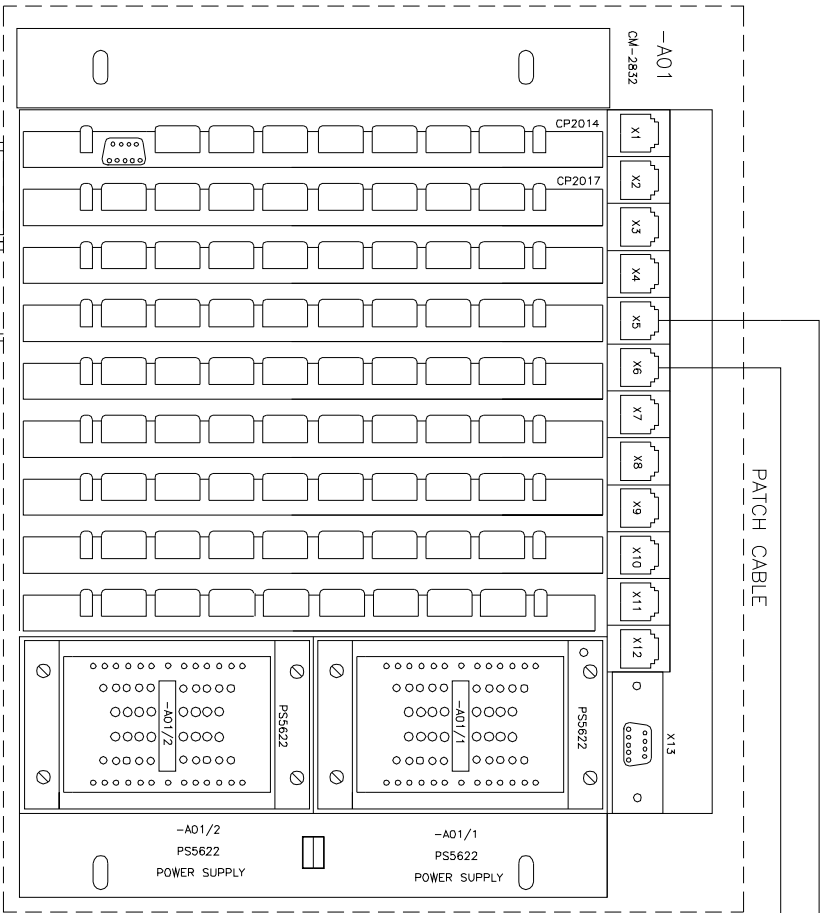
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Diese Zeichnung ist Eigentum der ANDRITZ HYDRO GmbH		Ges./Appr.		20.06.2016		Fuchs		VINH SON	
Perzingerstr./6 1141 Wien, ÖSTERREICH-Schritex, ISO 16016 beachten		Mossstab/Scale		-----		Lost Edit: 20.07.2016		Projec - No :	
Änderung / Modification		Datum/Date		Beorb./Des.		Mossstab/Scale		Projec - No :	

ANDRITZ		TURBINE CONTROL CUBICLE	
HYDRO		Circuit Diagram	
		Power Supply Cubicle Light, Socket Outlets, Cubicle Heater / Fan	

Int.No		= VINH SON	
Zech./Draw No.		+MET106H100	
Index		Bl./Sht. 12	
von/of 69		A3	



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			Gep./Check 20.06.2016 Fuchs		Anlage/Plant: VINH SON		
			Ges./Appr. 20.06.2016 Fuchs		Projct - No :		
			Mossstab/Sole		----- Lost Edit: 11.05.2012		
Änderung / Modification			Datum/Date		Beorb./Des.		



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Änderung / Modification				Datum/Date	Beorb./Des.				

ANDRITZ HYDROTECHNIK Hydro				TURBINE CONTROL CUBICLE				Int.No		= VINH SON	
Circuit Diagram								Zeich./Draw.No.		+MET100H100	
Control Unit AK 1703 ACP								Index		Bl./Sht. 14	
								von/of 69		A3	

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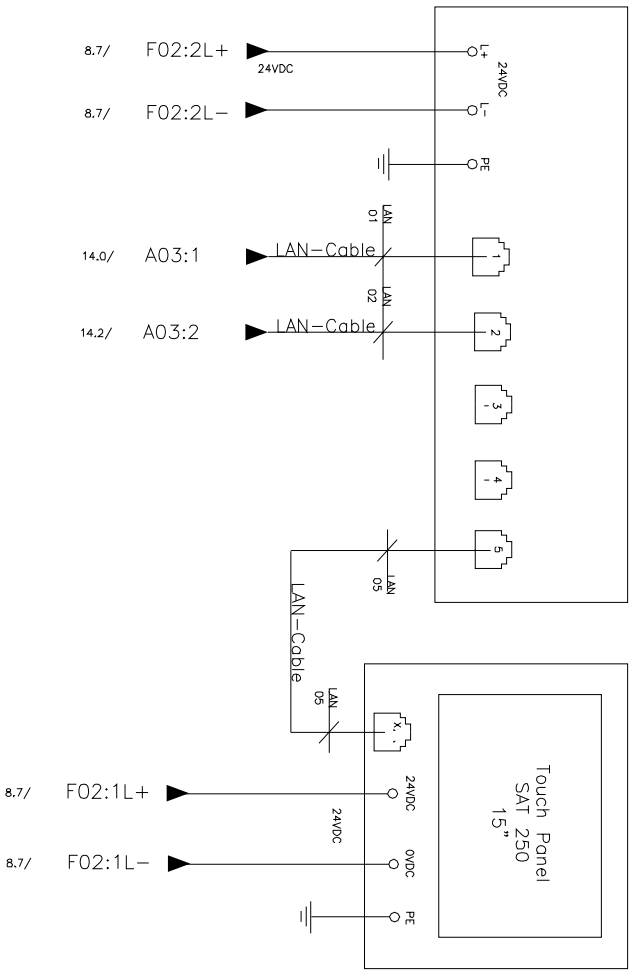
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—A03
Ethernet Switch
HIRSCHMANN
SPIDER 5TX

IP 172.XX.XXX.X1
—TP

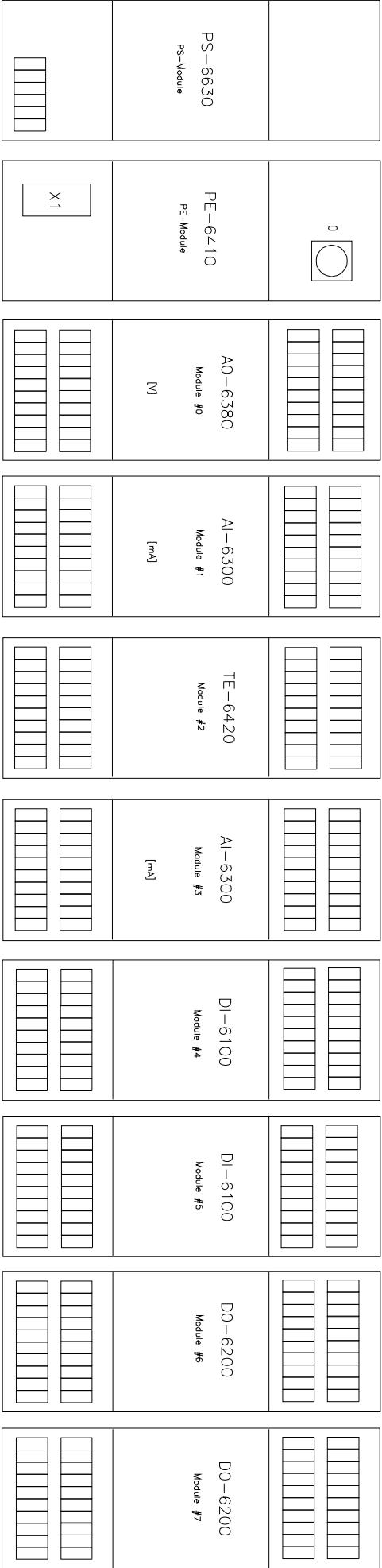


This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Patentreferat/76 1141 Vienna Austria-e.c., project, note ISO 16016 Diese Zeichnung ist GEISTIGES EIGENTUM der ANDRITZ HYDRO GmbH Patentreferat/76 1141 Wien, ÖSTERREICH-Schutzz. ISO 16016 beachten				Gez./Draw	20.06.2016	Auer	Kunde/Client:
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Änderung / Modification				Datum/Date	Bearb./Des.		

ANDRITZ Hydro				TURBINE CONTROL CUBICLE				InL.No		= VINH SON	
Circuit Diagram				Circuit Diagram				Zeich./Draw.No.		+MET/06H100	
Ethernet Switch, Touch Panel								Index		B1./Sht. 15	
								von/of 69		A3	

Terminal Module

Rail #0



Power Supply

Peripherie Controller (max. 8 Units)

4x Analog Output +/−10V

4x Analog Input 4−20mA

4x SMS (Speed)

4x Analog Input 4−20mA

16x Digital input

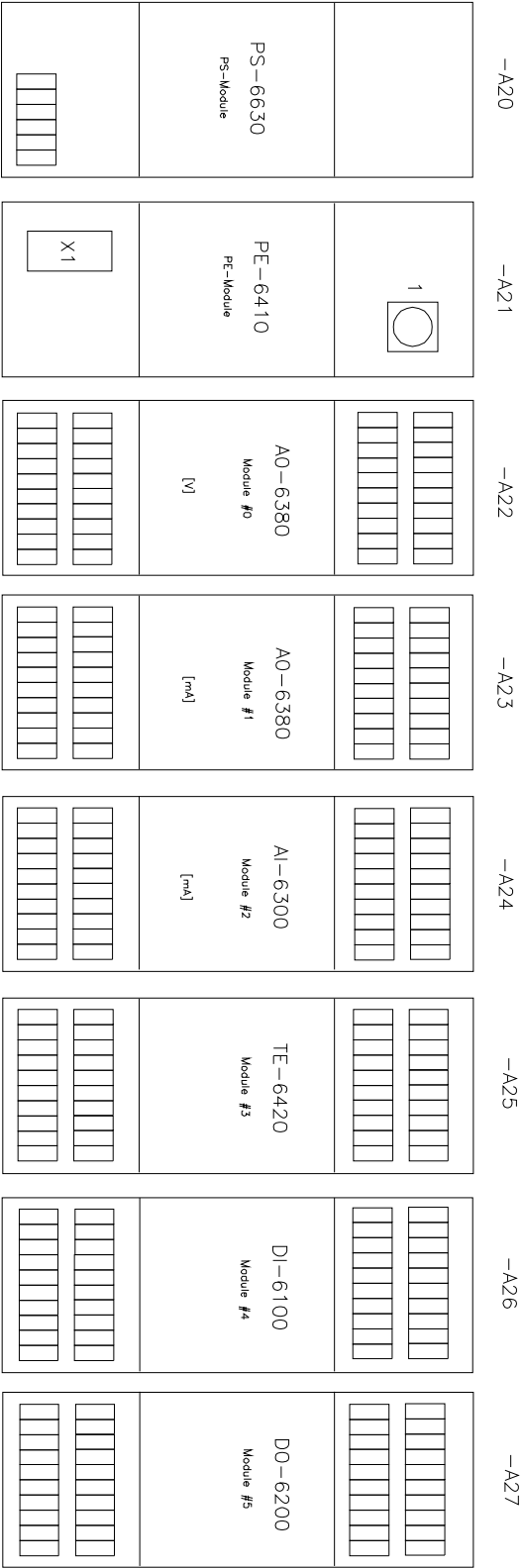
16x Digital Input

16x Digital Output

16x Digital Output

Terminal Module

Rail #1



Power Supply

Peripherie Controller (max. 8 Units)

4x Analog Output
+/-10V

4x Analog Output
+/-10V

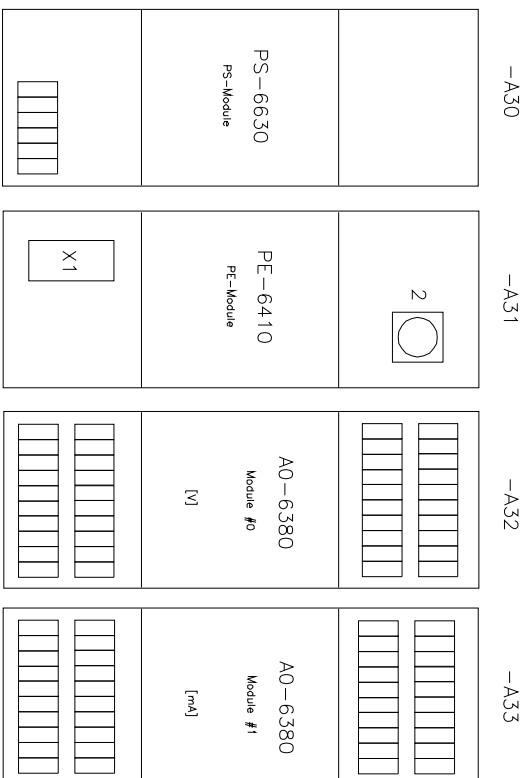
4x Analog Input
4-20mA

4x SMS (Speed)

16x Digital Input

16x Digital Output

Terminal Module
Rail #2



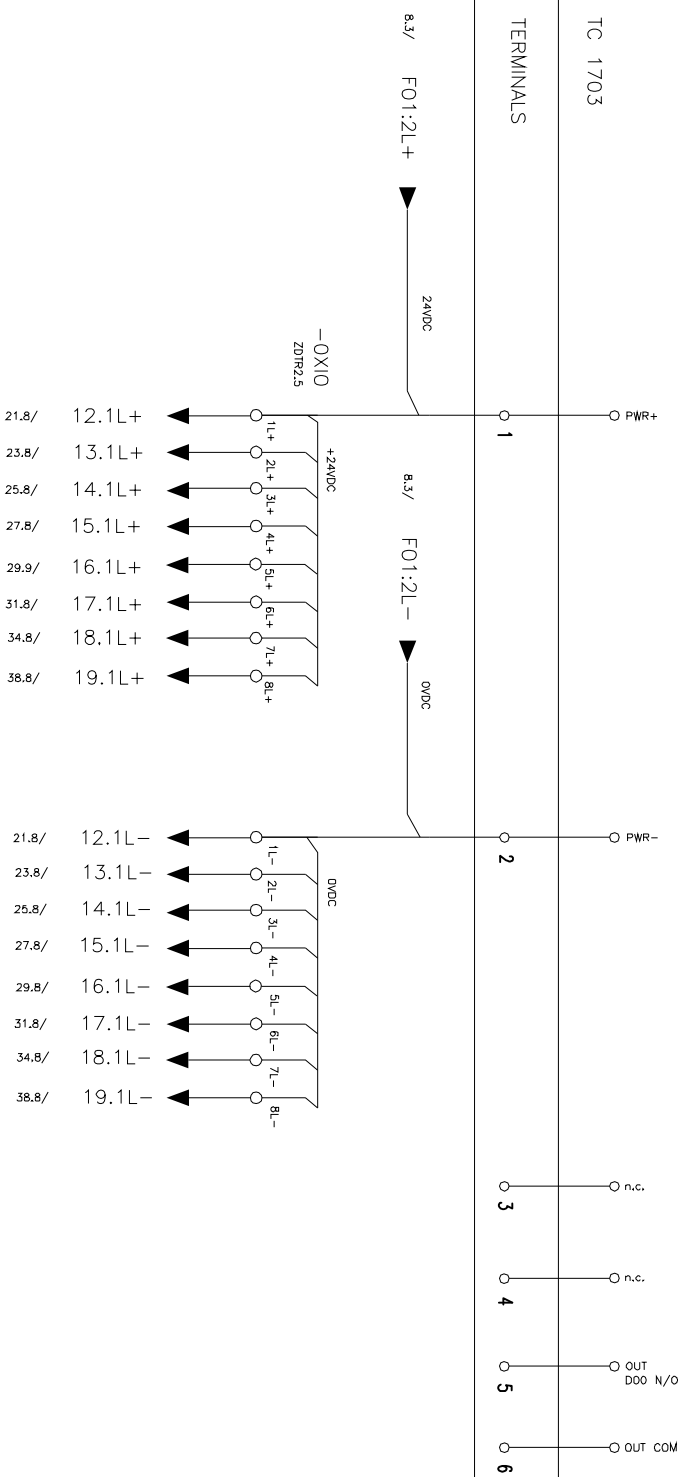
Power Supply	Peripherie Controller (max. 8 Units)	4x Analog Output +/−10V	4x Analog Output +/−10V

- A10

ADDRESS: PS 0, PS-MODULE
I/O-MODULE: PS-6630

TC 1703

TERMINALS



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Penzingerstr. 76 1141 Vienna Austria - A. project, note IS
Diese Zeichnung ist GEISTIGES EIGENTUM der ANDRITZ HY
Penzingerstr. 76 1141 Wien, ÖSTERREICH - Schulz, ISO 16

Gez./Draw	20.06.2016	Auer	Kunde/Client:
Gep./Check	20.06.2016	Fuchs	Anlage/Plant: VINH SON

	TURBINE CONTROL CUBICLE Circuit Diagramm
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Attachment / Meeting

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Book / Date

Model 1	
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VIN12-COV-0021

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ANDRITZ Hydro

TURBINE CONTROL CUBICLE
Circuit Diagramm
Power Supply #1 – Rail 0
Periphery

VIN12-CON-0021

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A3

–A11

ADDRESS: PE-MODULE
MODULE: PE-6401

TC 1703



14/7/ USB/1 

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				Mossstab/Scale	-----		Lost Edit:	20.07.2016		
Änderung / Modification				Datum/Date						
				Beorb./Des.						



ANDRITZ
Hydro

TURBINE CONTROL CUBICLE
Circuit Diagram
Peripherycontroller #1

Int.No

= VINH SON

Zeich./Draw.No.

Index

von/of 69

VN12-COV-0021

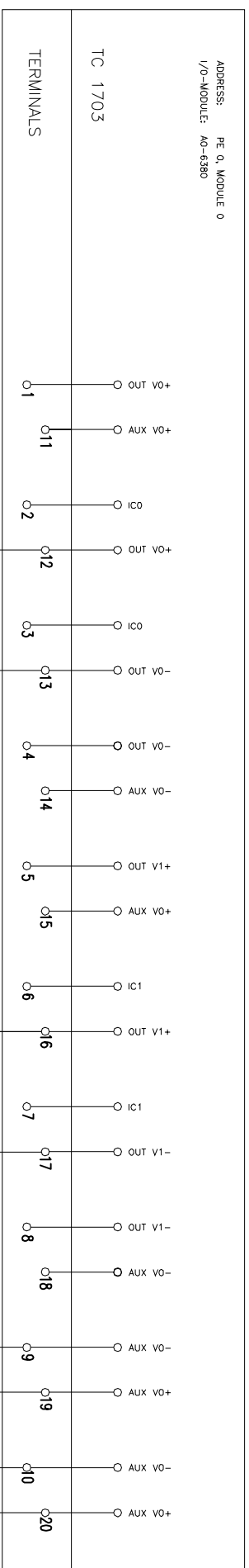
A3

-A12

ADDRESS: PE 0, MODULE 0
I/O-MODULE: AO-6380

Control	Signal	Needle
+/-10V	1	
-100%...+100%		
SIG-N1		

AO 1
+/-10V
Control Signal Deflector
-100%...+100%
SIG-DE1



Pinout diagram for the 16-pin DIP package of the F03.1L+ and F03.1L- comparators. The package is divided into two halves by a diagonal line from pin 1 to pin 8. The left half (pins 1-7) is for the F03.1L+ and the right half (pins 8-16) is for the F03.1L-.

Pin	Function
1	24VDC
2	0-10V
3	0-6A
4	0-10V
5	0-6A
6	0-10V
7	0-6A
8	24VDC
9	0-10V
10	0-6A
11	0-10V
12	0-6A
13	0-10V
14	0-6A
15	0-10V
16	0-6A

[illegible]

	/22.8	12.2L-	▶
	/22.8	12.2L+	▶
24VDC	/19.4	12.1L-	▶
	/19.2	12.1L+	▶

INTERNAL

EXTERNAL

NEEDLE 1
ACTUATOR

32W
12Vdc

ACTUATOR
TYPE ED12

PE

+XXX-S01

NEVPIC - ELECTRO - HYDR
Transducer ED12

ACTUATOR
1
2
PE
+
-
ACTUATOR
TYPE ED25
32W
12Vdc
EXISTING
+XXX-S02
NEYRIC - ELECTRO - HYDR
Transducer ED25

MM	Änderung / Modification
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66016	Gez./Draw	20.06.2016	Auer	Kunde/Client:
20 cm1	Gepl./Check	20.06.2016	Fuchs	Analoge/Plantt:
beachten	Gas./Appr.	20.06.2016	Fuchs	Projec - Nr. :
	Mosastab/Sole	-----	Lost Edit: 11	
Datum/Date	Berch./Des.			

	VINH SON
	05.2012

ANDRITZ
Hydro

TURBINE CONTROL CUBICLE
Circuit Diagram
Roll #0
AD-6380 Module 0

Int.No	= VINH SON		
	+ME10GH100		
Zeich./Draw.No.	Index	B1./Sht. 21 von/ of 69	A3
VIN12-GOV-0021			

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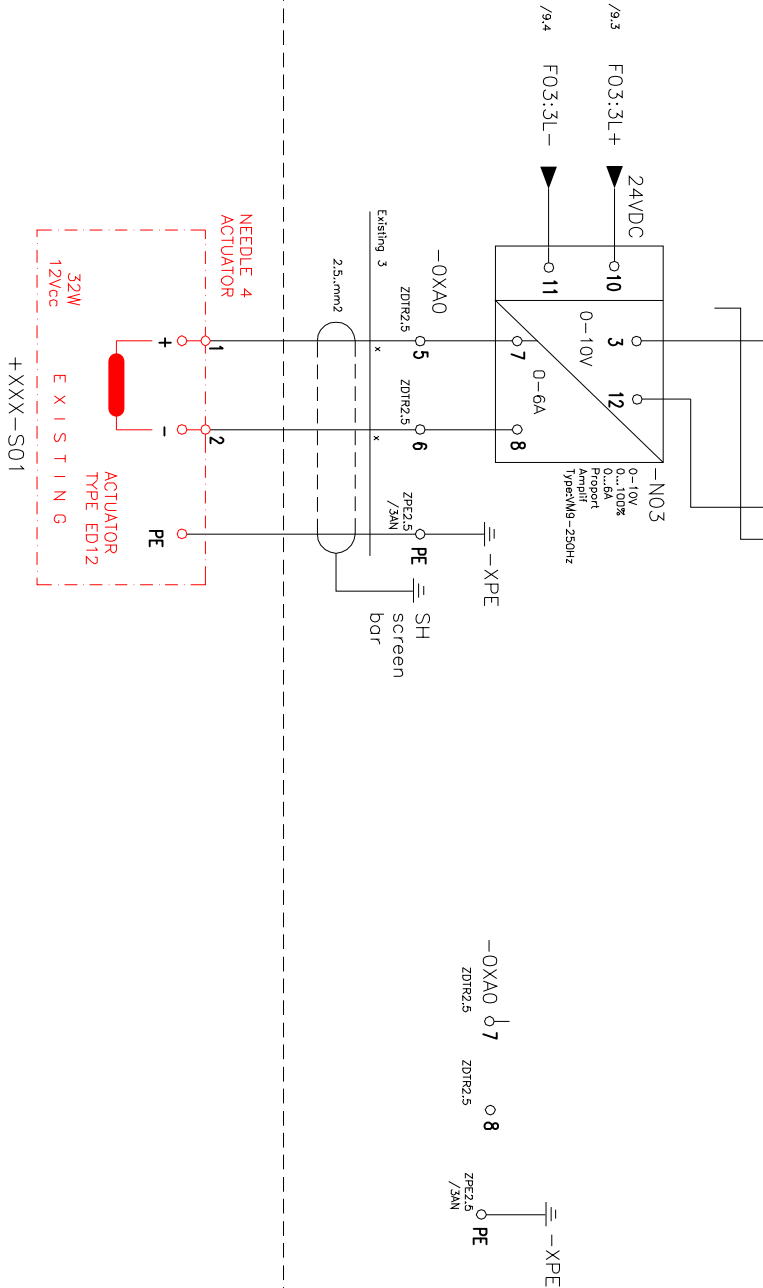
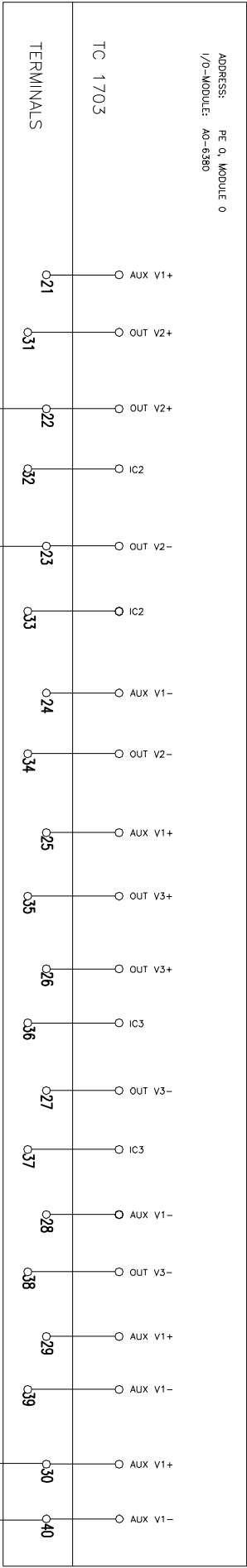
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-A12

AO 2
+/-10V
Control Signal Needle 4
-100%...+100%
SIG-N1

AO 3
+/-10V
S P A R E



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				Gez./Draw	20.06.2016	Auer	Kunde/Client:								
				Geop./Check	20.06.2016	Fuchs	Anlage/Plant:	VINH SON							
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Änderung / Modification				Datum/Date	Beorb./Des.										

ANDRITZ Hydro				TURBINE CONTROL CUBICLE				InL.No				= VINH SON			
Rail #0				Circuit Diagramm				+MET10GH100				Index			
AD-6380				Module 0				Zeich./Draw.No.				B1./Sht. 22			
								VIN2-60V-0021				von/of 69			
												A3			



ANDRITZ
HYDRO
Turbine Control Cubicle
Circuit Diagram
Reil #0
AO-6380 Module 0

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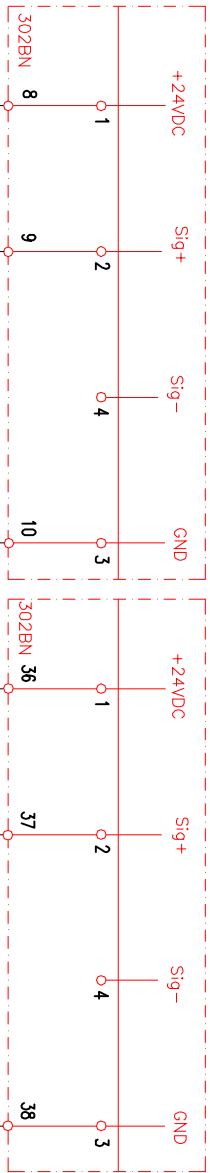
9

AI 0
4–20mA
Feedback Signal Needle 1
0,...100%
N1–SIGNAL

AI 1
4–20mA
Feedback Signal Deflector
0,...100%
DE1–SIGNAL

02GTA002CR

02GTA002CR

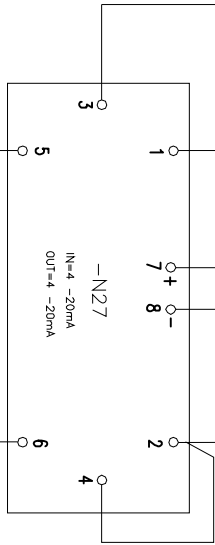


EXTERNAL

INTERNAL

F04:2L– /9:7
F04:2L+ /9:7

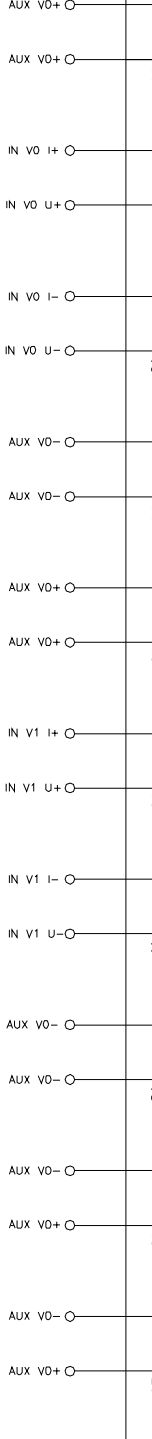
N28:8– /24:3
N28:7+ /24:3



–A13

13.2L– /24,8
13.2L+ /24,8

TERMINALS



ADDRESS: PE 0, MODULE 1
I/O–MODULE: AI–6300

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				Gepr./Check 20.06.2016		Anlage/Plant: VINH SON		+METROCH100	
				Ges./Appr. 20.06.2016		Projec - No : Fuchs		Zeich./Draw.No. VIN12-COV-0021	
Änderung / Modification				Datum/Date		Beorb./Des.		Index	
				Mossstab/Sole		Lost Edit: 11.05.2012		BT./Sht. 23 von/of 69	
								A3	

ANDRITZ

Hydro

TURBINE CONTROL CUBICLE		= VINH SON	
Circuit Diagramm		+METROCH100	
Roll #0	Module 1		
Al-6300			



Hydro

TURBINE CONTROL CUBICLE

Circuit Diagramm

Roll #0

AI–6300 Module 1

0

1

2

3

4

5

6

7

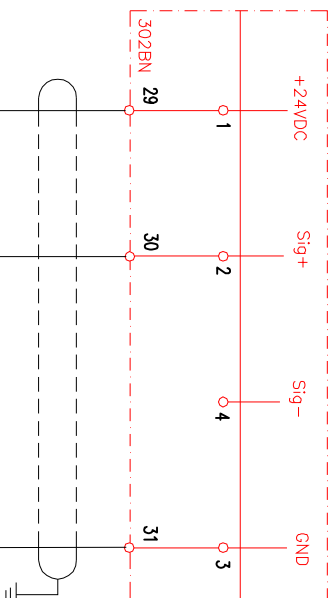
8

9

AI 2
4-20mA
Feedback Signal Needle 4
0...100%
N4-SIGNAL

AI 3
4-20mA
SPARE

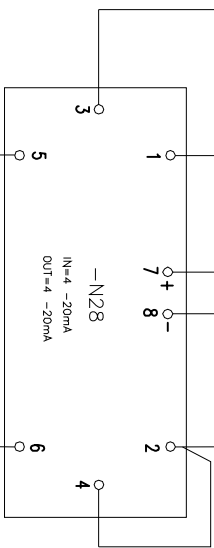
02GTA002CR



EXTERNAL

INTERNAL

N27:8- /23,5
N27:7+ /23,5
N29:8- /43,1
N29:7+ /43,1



-A1 3

-OXA1
ZDR2,5

φ12

φ13

φ15

φ16

φ14

TERMINALS

TC 1703

ADDRESS: PE 0, MODULE 1
I/O-MODULE: AI-6300

AUX VO+

AUX VO+

IN VO +

IN VO +

IN VO -

IN VO -

AUX VO-

AUX VO-

AUX VO+

AUX VO+

IN V1 +

IN V1 +

IN V1 -

IN V1 -

AUX VO-

AUX VO-

AUX VO-

AUX VO+

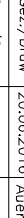
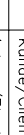
AUX VO-

AUX VO+

13.2L- /23,8

13.2L+ /23,8

This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Patentreferenz: 1141 Vienna Austria-c.l. project, note ISO 16016 Dieses Zeichnung ist EIGENTUM der ANDRITZ HYDRO GmbH Patentreferenz: 1141 Wien, ÖSTERREICH-Schutz. ISO 16016 beachten									
		Gez./Draw	20.06.2016	Auer	Kunde/Client:				
		Gepr./Check	20.06.2016	Fuchs	Anlage/Plant:	VINH SON			
		Ges./Appr.	20.06.2016	Fuchs	Projec - No :				
		Mossstab/Sole	-----						
Anderung / Modification		Datum/Date	Beorb./Des.		Lost Edit: 11.05.2012				

TURBINE CONTROL CUBICLE		Int.No		= VINH SON	
Circuit Diagramm		+HE100H100			
Rad #0		Zeich./Draw.No.		Index	
Al-6300		VIN2-C0V-0021		B1./Snt. 24	
Module 1				von/of 69	
				A3	

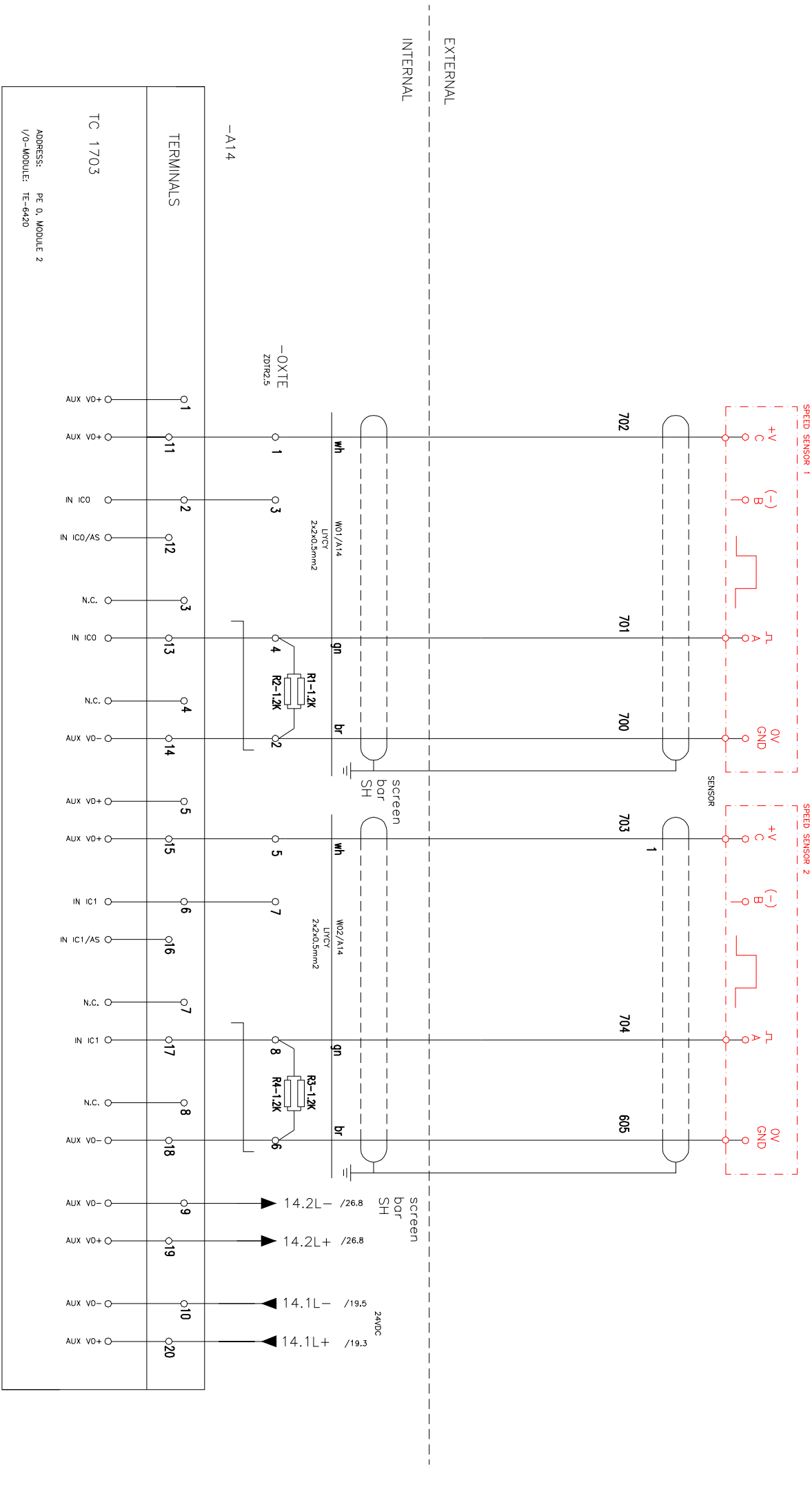
ANDRITZ Hydro				TURBINE CONTROL CUBICLE				Int.No		= VINH SON	
Raid #0				Circuit Diagramm				Zech./Draw.No.		+METROCH100	
AI-6300 Module 1								Index		Bl./Sht. 24	
								von/of 69		A3	

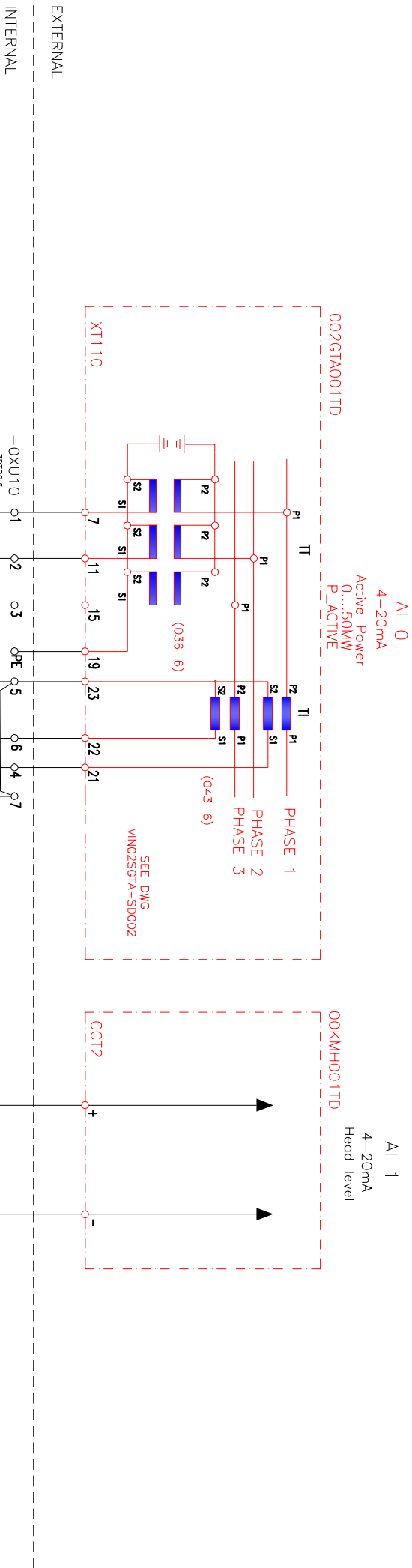
SMS 0

Speed Signal From Pick up 1
n1-SMS

SMS 1

Speed Signal From Pick up 2
n3-SMS



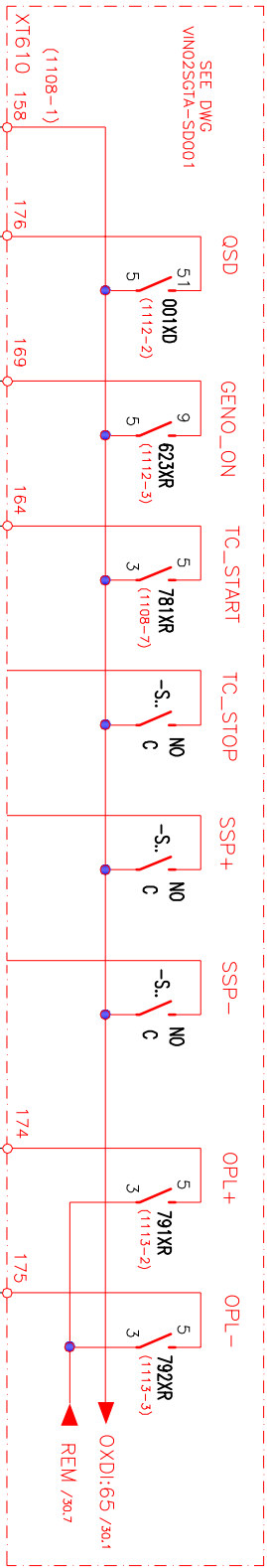


This drawing is the property of ANDRITZ HYDRO GmbH Perzogenstr./26 1141 Vienna Austria-e.c. project, note ISO 16016 Dieses Zeichnung ist GESAMTES EIGENTUM der ANDRITZ HYDRO GmbH Perzogenstr./26 1141 Wien, OSTERREICH-Schuttr. ISO 16016 beachten				Gez./Draw 20.06.2016 Auer		Kunde/Client: VINH SON		ANDRITZ Hydro		TURBINE CONTROL CUBICLE		Int.No		= VINH SON	
				Geop./Check 20.06.2016 Fuchs		Anlage/Plant: VINH SON				Circuit Diagramm		Zeich./Draw.No.		+METROCH100	
				Ges./Appr. 20.06.2016 Fuchs		Projec - No :				Rail #0		Index		Bl./Sht. 27	
				Mossstab/Sole		Lost Edit: 11.05.2012				AI-6300 Module 3		VIN2-COV-0021		von/of 69	
Änderung / Modification				Datum/Date		Beorb./Des.								A3	

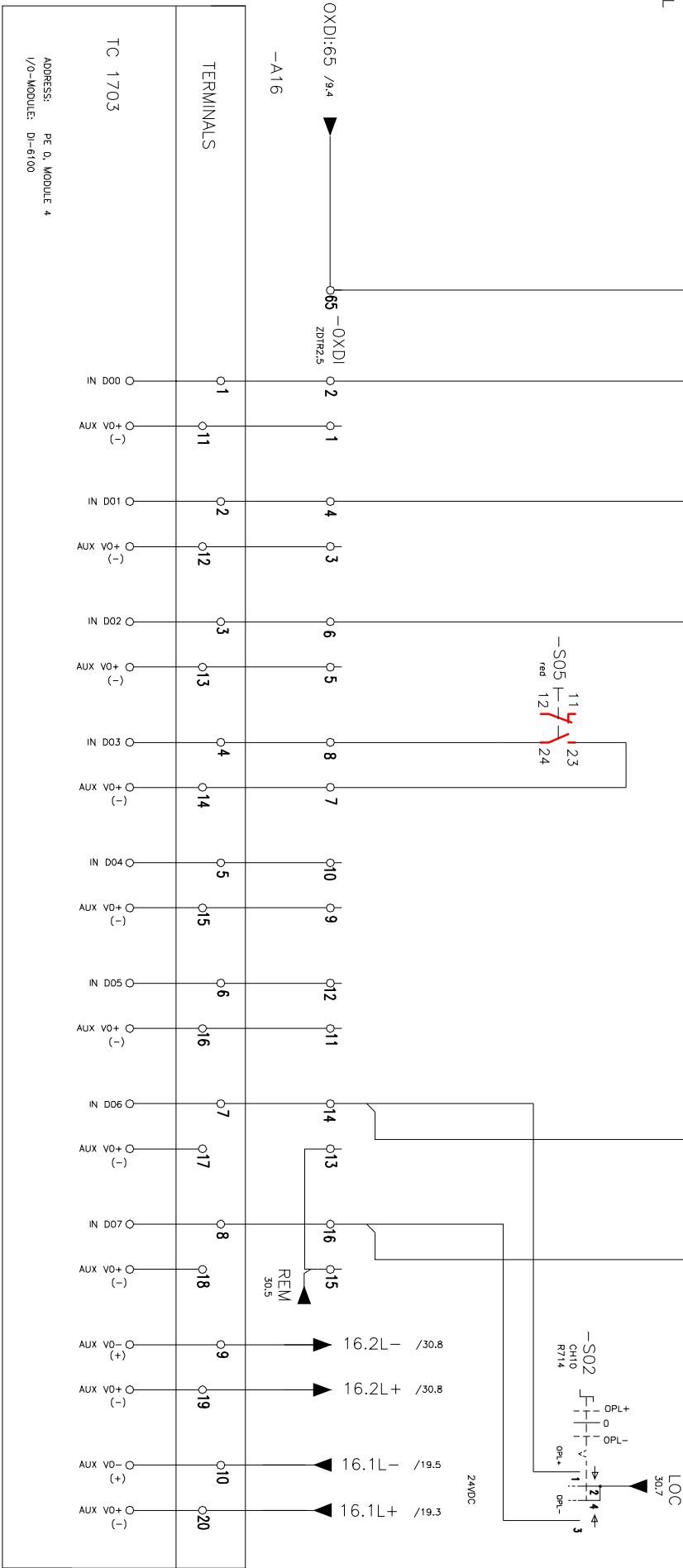
- DI 0DI 1DI 2DI 3DI 4DI 5DI 6DI 7
- Quick shut downCircuit Breaker ClosedTurbine Controller StartPanel ShutdownSpeed Setpoint HigherSpeed Setpoint LowerOpening Limiter HigherOpening Limiter Lower

02GTA003AR

SEE DWG
VIN02SGTA-SD001



EXTERNAL
INTERNAL

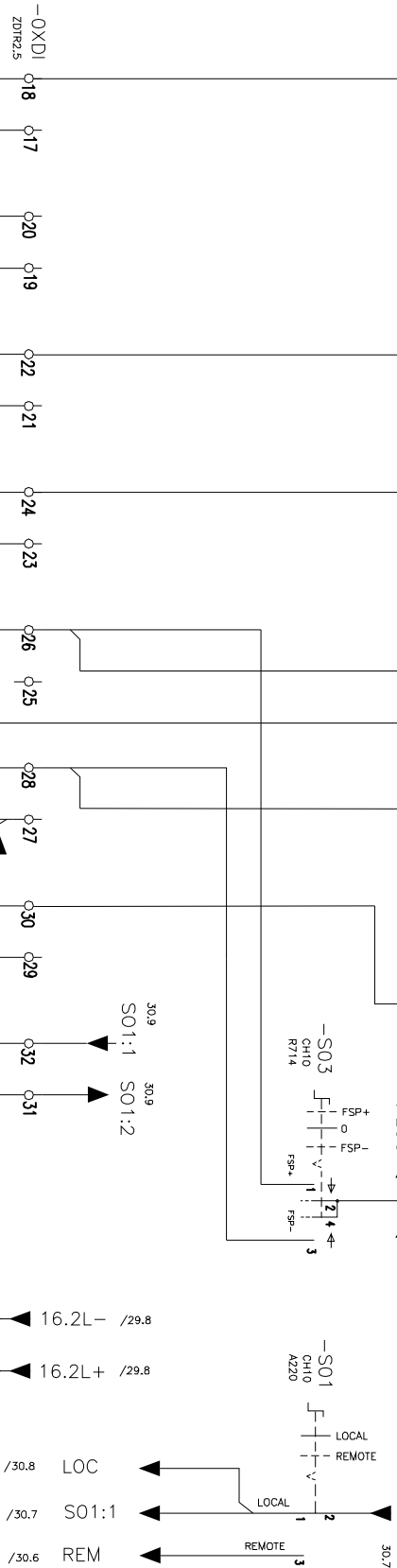
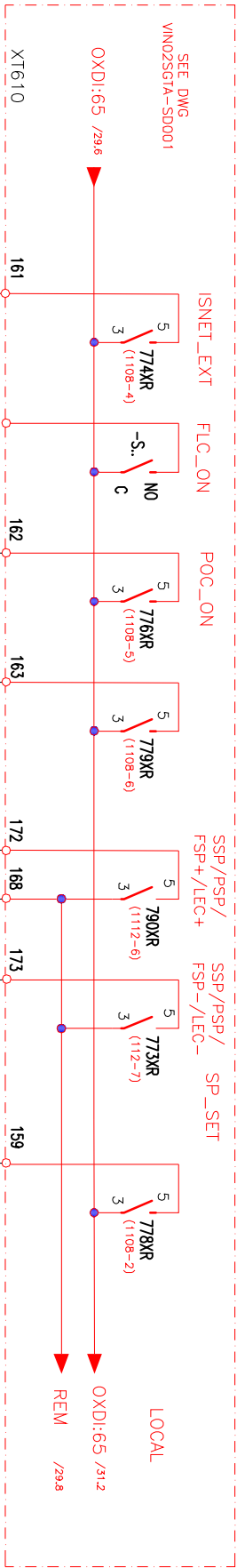


ADDRESS: PE 0, MODULE 4
I/O-MODULE: DI-6100

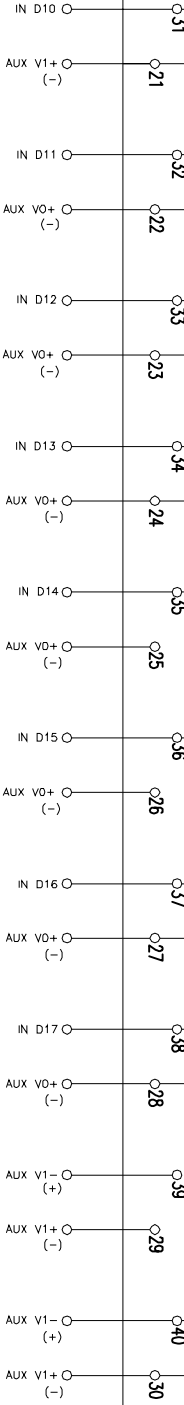
- DI 10 DI 11 DI 12 DI 13 DI 14 DI 15 DI 16 DI 17
- Droop change BP0/01 Flower Controller on Power Controller on SPARE Setpoint Higher Setpoint Lower Set Load Setpoint enable setpoint FSP/PSP Turbine Controller Local Mode

02GTA003AR

SEE DWG
VIN02SGTA-SD0001



ADDRESS: PE 0, MODULE 4
I/O-MODULE: DI-6100



- DI 0Manual Synchron On
- DI 1Primary Control On
- DI 2Forcing Permanent At no load parameter
- DI 3Speed No Load De Load Default to 0 MW
- DI 4Line Charging
- DI 5SPARE
- DI 6Reset after turbine Governor Trip
- DI 7De-pressurize Ring Pipe

02GTA003AR

PC_ON

RESET N_DISCHARG

SEE DWG
VINQ2SGTA-SD001

OXDI:65 /30.7

789XR
(1112-5)

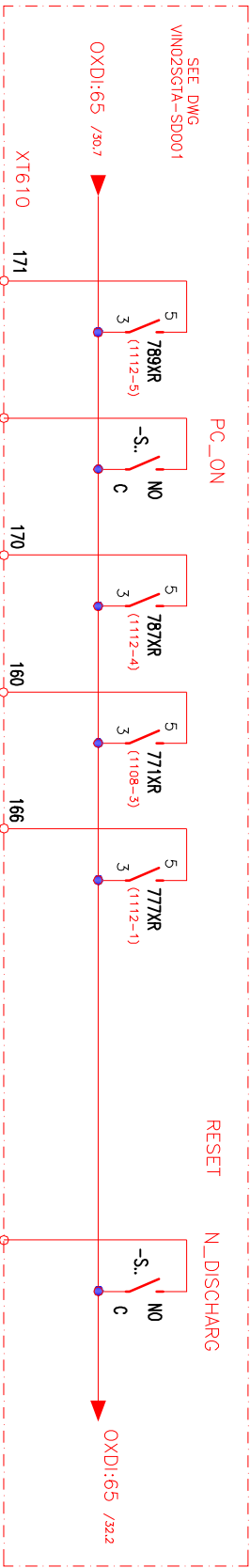
-S..
NO
C

787XR
(1112-4)

771XR
(1108-3)

777XR
(1112-1)

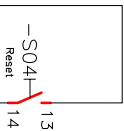
OXDI:65 /32.2



EXTERNAL
INTERNAL

-A17

-OXDI
ZDTR2.5



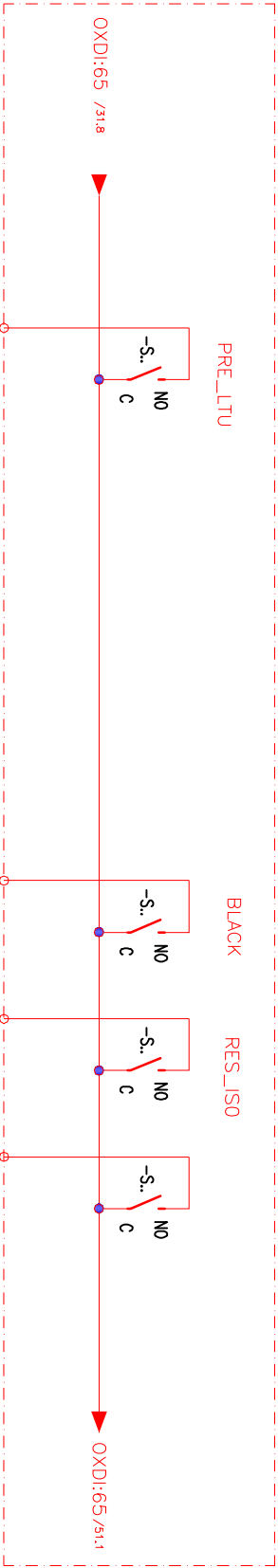
24VDC

TERMINALS		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
IN D00	○	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
AUX V0+ (-)	○	11	12	13	14	15	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20
IN D01	○	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
AUX V0+ (-)	○	11	12	13	14	15	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20
IN D02	○	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
AUX V0+ (-)	○	13	14	15	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20		
IN D03	○	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			
AUX V0+ (-)	○	14	15	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20			
IN D04	○	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20				
AUX V0+ (-)	○	15	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20				
IN D05	○	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20					
AUX V0+ (-)	○	16	17	18	19	20	11	12	13	14	15	16	17	18	19	20					
IN D06	○	7	8	9	10	11	12	13	14	15	16	17	18	19	20						
AUX V0+ (-)	○	17	18	19	20	11	12	13	14	15	16	17	18	19	20						
IN D07	○	8	9	10	11	12	13	14	15	16	17	18	19	20							
AUX V0+ (-)	○	18	19	20	11	12	13	14	15	16	17	18	19	20							
AUX V0+ (-)	○	9	10	11	12	13	14	15	16	17	18	19	20								
AUX V0+ (-)	○	19	20	11	12	13	14	15	16	17	18	19	20								
AUX V0+ (-)	○	20	11	12	13	14	15	16	17	18	19	20									
AUX V0+ (-)	○	20	11	12	13	14	15	16	17	18	19	20									

ADDRESS: PE 0, MODULE 5
I/O-MODULE: DI-6100

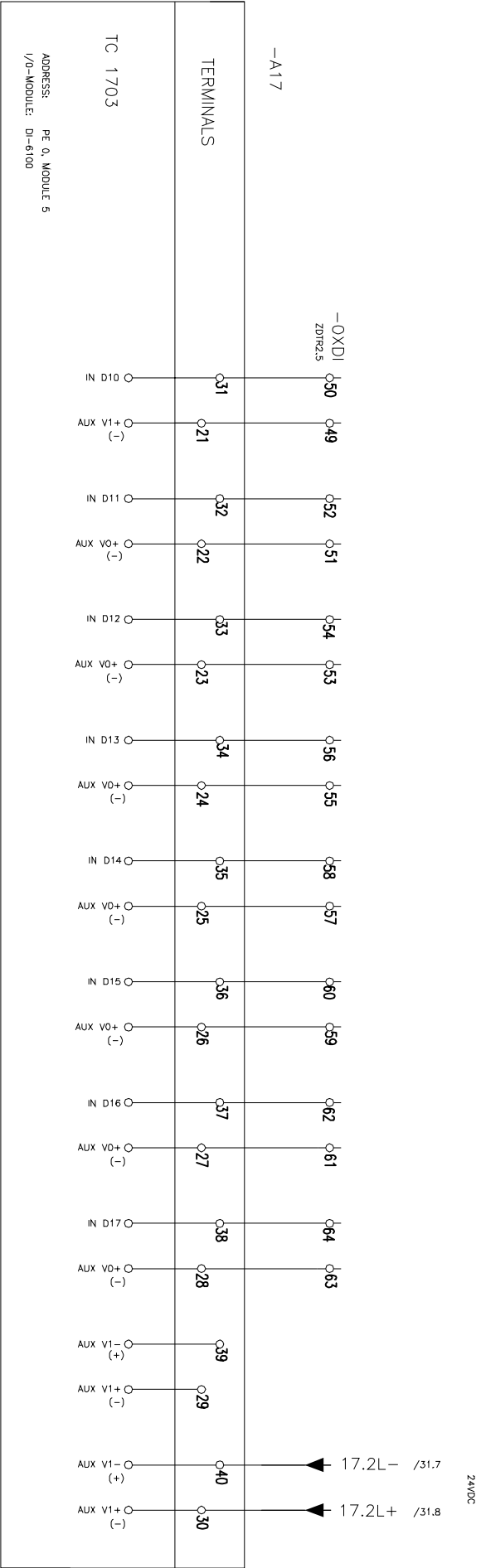
TC 1703

DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 7
Preparation for Load Take Up	SPARE	SPARE	SPARE	Block- Start	Reset Inland Operation	Nozzle Freezing	SPARE



EXTERNAL

INTERNAL

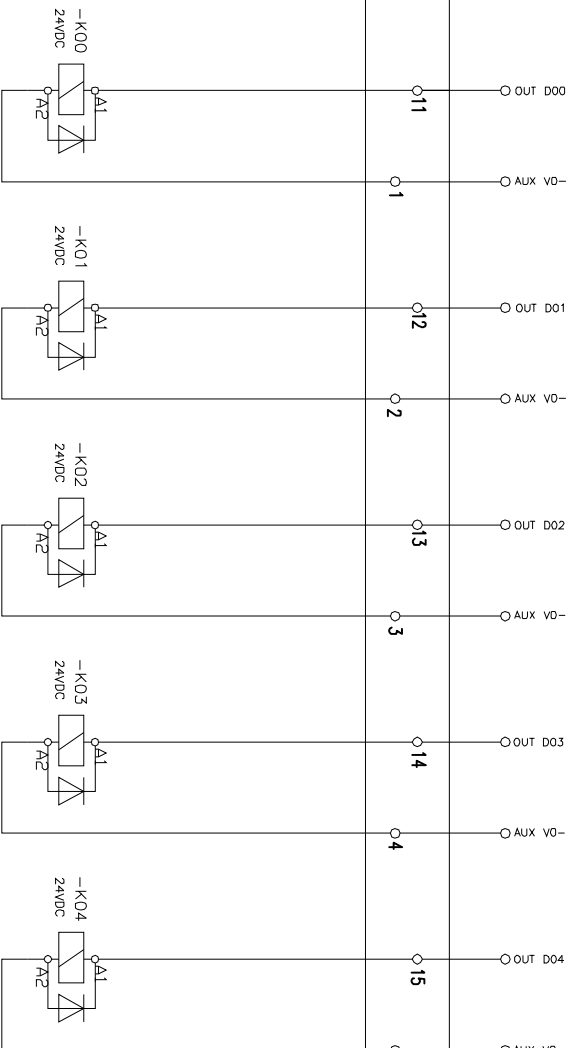


-A18	DO 0 Major Fault No Turbine Governor Shut Down NOT-ESD-QSD	DO 1 Minor Fault General Alarm FAIL-IT	DO 2 Turbine Limited TU-LIMITED	DO 3 Isolating Operation Active ACTIVE	DO 4 Forcing parameter at no load
------	--	---	---------------------------------------	---	---

ADDRESS: PE 0, MODULE 6
I/O-MODULE: DO-6200

TC 1703

TERMINALS



INTERNAL

EXTERNAL

Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24
14 11 63.1 12 24 21 63.2 22 34 31 32 44 41 42	14 11 63.4 12 24 21 63.5 22 34 31 32 44 41 42	14 11 63.5 12 24 21 63.6 22 34 31 32 44 41 42	14 11 63.8 12 24 21 63.9 22 34 31 32 44 41 42	

-A18

DO 5

PRIORITY NEEDLE 2

FLC_ON_INT

DO 6

PRIORITY NEEDLE 3

POC_ON_INT

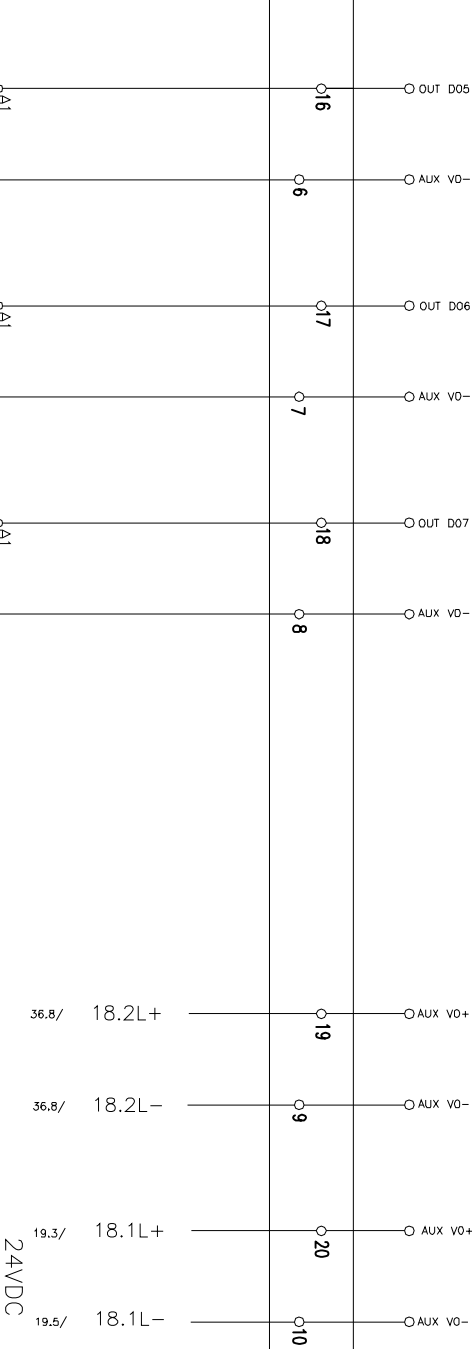
DO 7

TC Ready To Start

ADDRESS: PE 0, MODULE 6
I/O-MODULE: DO-6200

TC 1703

TERMINALS

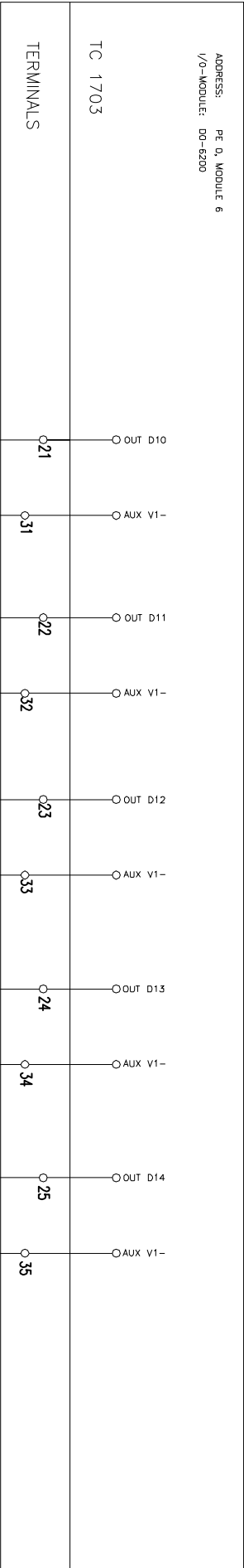


INTERNAL

EXTERNAL

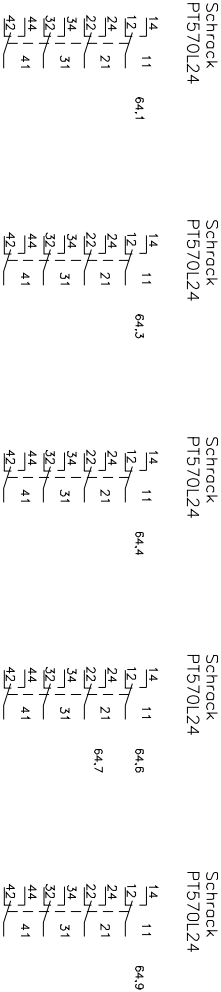
Schrack PT570L24	Schrack PT570L24	Schrack PT570L24
14 11	14 11	14 11
12 11	12 11	12 11
24 21	24 21	24 21
22 21	22 21	22 21
34 31	34 31	34 31
32 31	32 31	32 31
44 41	44 41	44 41
42 41	42 41	42 41
63.1	63.3	63.4

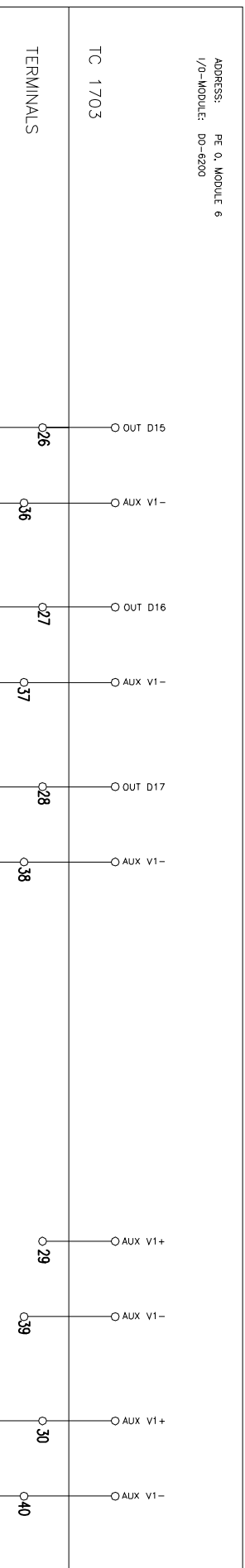
	DO 10	DO 11	DO 12	DO 13	DO 14
	Speed Switch	Speed Switch Brake	Speed Switch Standstill	Speed Switch Brake	Speed Switch Excitation
-A18	n<80%	n<10%	n<1%	n<15%	n<70%



INTERNAL

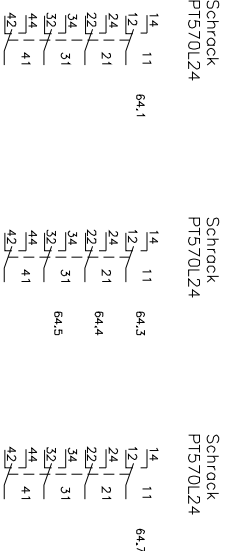
EXTERNAL



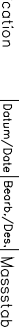


INTERNAL

EXTERNAL



This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Penzbergstr./76 1141 Velden Austria, project, note ISO 16016 Diese Zeichnung ist EIGENTUM der ANDRITZ HYDRO GmbH Penzbergstr./76 1141 Velden, ÖSTERREICH-Schulz, ISO 16016 beachten		Gez./Draw 20.06.2016 Auer		Kunde/Cient: 20.06.2016 Auer	
Datum/Date 20.06.2016 Fuchs		Gez./Check 20.06.2016 Fuchs		Anlage/Plant: VINH SON	
Änderung / Modification		Ges./Appr. 20.06.2016 Fuchs		Project - No : Lost Edit: 11.05.2012	
Massstab/Scale		-----		-----	



TURBINE CONTROL CUBICLE Circuit Diagramm		Int.No = VINH SON +ME70GH100	
Zeich./Draw.No. VIN2-G0V-0021		Index Bl./Sh. 36 von/of 69	
Rail 10 DO-6200 Module 6		A3	

-A19

DO O
spare

DO 1
Needle 2
not bloked

DO 2
Needle 3
not bloked

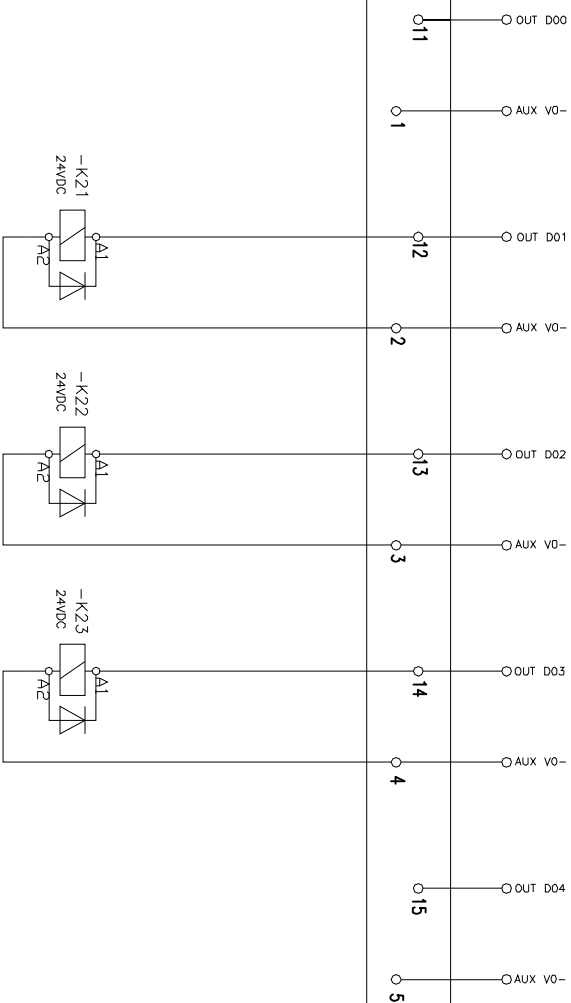
DO 3
Needle 4
not blocked

DO 4
spare

ADDRESS: PE 0, MODULE 7
I/O-MODULE: DO-6200

TC 1703

TERMINALS



INTERNAL

EXTERNAL

Schrock PT570L24	Schrock PT570L24	Schrock PT570L24
$\frac{14}{12}$ $\frac{11}{21}$ $\frac{24}{22}$ $\frac{31}{41}$ $\frac{44}{42}$ $\frac{41}{22}$ $\frac{42}{22}$ $\frac{41}{21}$ 65.3	$\frac{14}{12}$ $\frac{11}{21}$ $\frac{24}{22}$ $\frac{31}{41}$ $\frac{44}{42}$ $\frac{41}{22}$ $\frac{42}{22}$ $\frac{41}{21}$ 65.4	$\frac{14}{12}$ $\frac{11}{21}$ $\frac{24}{22}$ $\frac{31}{41}$ $\frac{44}{42}$ $\frac{41}{22}$ $\frac{42}{22}$ $\frac{41}{21}$ 65.6

Schrack
Pf570124

Number of Children	Percentage (%)
1	65.4
2	11
3	12
4	24
5	21
6	22
7	34
8	31
9	32
10	44
11	41
12	42
13	41
14	42

Schrack
Pf570124

Number of Children	Percentage (%)
1	65.6
2	11
3	12
4	24
5	21
6	22
7	34
8	31
9	32
10	44
11	41
12	42
13	41
14	42

Scrnack
PT570L24

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MM	Änderung / Modification

16016 0 GmbH 5 Beachten	Gez./Draw	20.0
	Gep./Check	20.0
	Ges./Appr.	20.0
Datum/Date	Beort./Des.	Mosstab/Scale

Gez./Draw	20.06.2016	Auer
Gepr./Check	20.06.2016	Fuchs
Gas./Apr.	20.06.2016	Fuchs
Masstab/Sale		-----

Kunde/Client:	
Anlage/Plant:	VINH SON
Projec - No :	
Last Edit:	20.07.2016

ANDRITZ
PUMPENSYSTEME
Hydro

TURBINE CONTROL CUBICLE
Circuit Diagram
Rail #0
DO-6200 Module 7

Inl.No	= VINH SON		
	+MET10GH100		
Zeich./Draw.No.	Index	B1./Sht. 37 von/of 69	A3

0

1

2

3

4

5

6

7

8

9

DO 5

Open
Limiter
at Max

DO 6

Power Less
Than 1MW

DO 7

Governor in
Automatic Mode

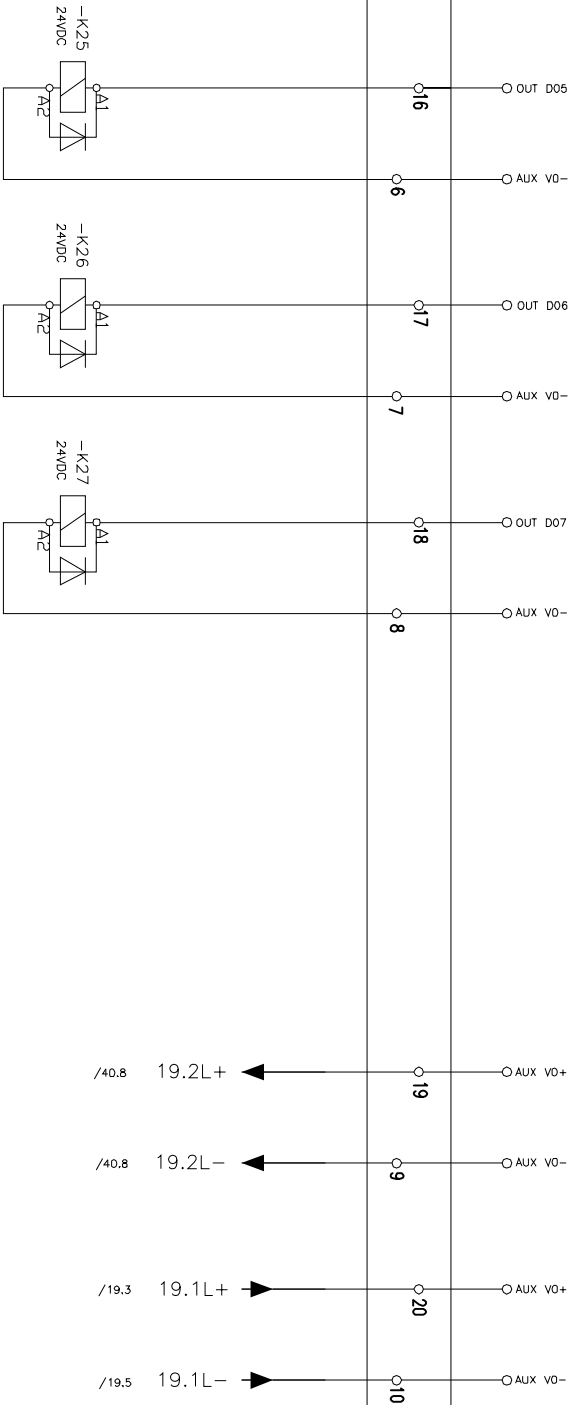
P<1MW

AUTO

ADDRESS: PE 0, MODULE 7
I/O-MODULE: DO-6200

TC 1703

TERMINALS



INTERNAL

EXTERNAL

Schrack
PT570L24

Schrack
PT570L24

Schrack
PT570L24

This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Persönlicher/76 1141 Vienna Austria-rl, project, note ISO 16016 Diese Zeichnung ist EIGENTUM der ANDRITZ HYDRO GmbH Persönlicher/76 1141 Wien, ÖSTERREICH-Schritex, ISO 16016 beachten				Gez./Draw	20.06.2016	Auer	Kunde/Client:	TURBINE CONTROL CUBICLE				InL.No	= VINH SON	
				Gepr./Check	20.06.2016	Fuchs	Anlage/Plant:	VINH SON				+METROCH100		
				Ges./Appr.	20.06.2016	Fuchs	Projec - No :	R01 #0				Zeich./Draw.No.	VIN12-CON-0021	
				Mossstab/Sole	-----		Lost Edit:	11.05.2012				Index	Bl./Sht. 38	
Änderung / Modification				Datum/Date									von/of 69	
				Beerb./Des.									A3	



Circuit Diagramm

DO 10

DO 11

DO 12

DO 13

DO 14

spore

spore

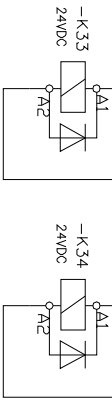
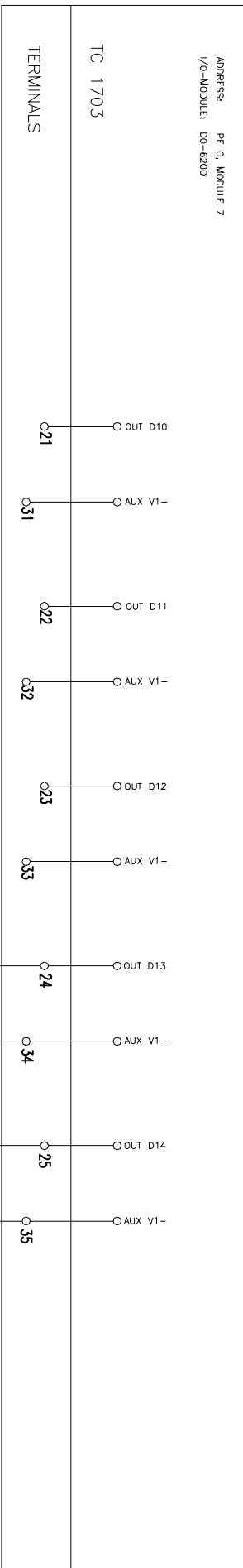
spore

Speed Signal Faulty

load frequency to zero position

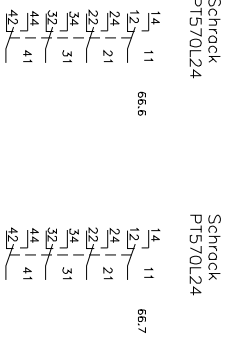
—A19

ADDRESS: PE 0, MODULE 7
I/O—MODULE: DO—6200



INTERNAL

EXTERNAL



D0 15

Start
LO_ON
feedback

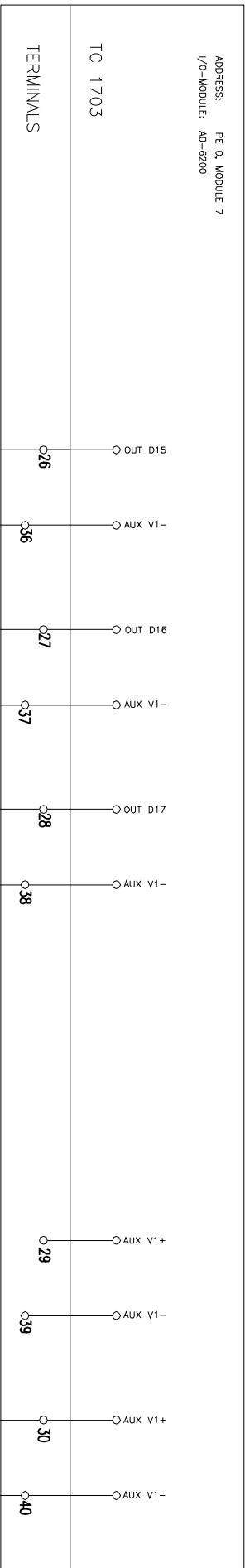
D0 16

PRIORITY NEEDLE 4
PC_ON

D0 17

Nozzles
Are Closed
N=0%

ADDRESS: PE 0, MODULE 7
I/O-MODULE: AO-6200



INTERNAL

EXTERNAL

Schrock	Schrock	Schrock
PT570L24	PT570L24	PT570L24
14	14	14
11	11	11
66.1		66.4
12	12	12
24	24	24
21	21	21
22	22	22
34	34	34
31	31	31
32	32	32
44	44	44
41	41	41
42	42	42

0

1

2

3

4

5

6

7

8

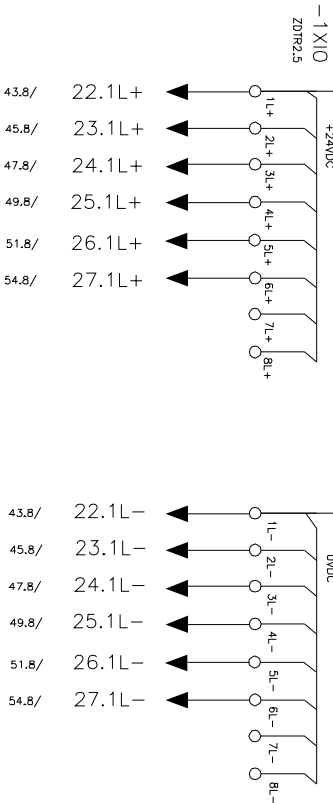
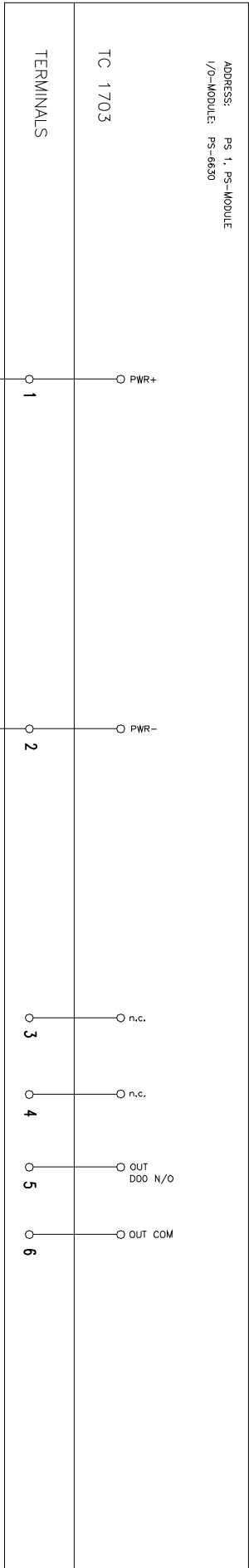
9

—A20

ADDRESS: PS 1, PS-MODULE
I/O-MODULE: PS-0630

TC 1703

TERMINALS



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				Gepr./Check 20.06.2016 Fuchs		Anlage/Plant: VINH SON						+METROCH100			
				Ges./Appr. 20.06.2016 Fuchs		Projec – No :						Index		Bl./Sht. 41	
ÄM	Änderung / Modification			Datum/Date	Beorb./Des.	Mosslab/Sole	Lost Edit: 20.07.2016		VIN12-CON-0021		von/of 69		A3		

–A21

ADDRESS: PE 1, PE-MODULE
MODULE: PE-6410

TC 1703



14.7/ USB/2

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				Gepr./Check 20.06.2016 Fuchs				Anlage/Plant: VINH SON			
				Ges./Appr. 20.06.2016 Fuchs				Projec – No :			
				Mossstab/Sole				----- Lost Edit: 20.07.2016			
Änderung / Modification				Datum/Date		Beorb./Des.					

ANDRITZ
Hydro

TURBINE CONTROL CUBICLE
Circuit Diagram
Peripherycontroller #2 – Rail 1

Int.No	= VINH SON	
Zeich./Draw.No.	VIN2-60V-0021	
Index	Bl./Sht.	42
von/of		69

A3

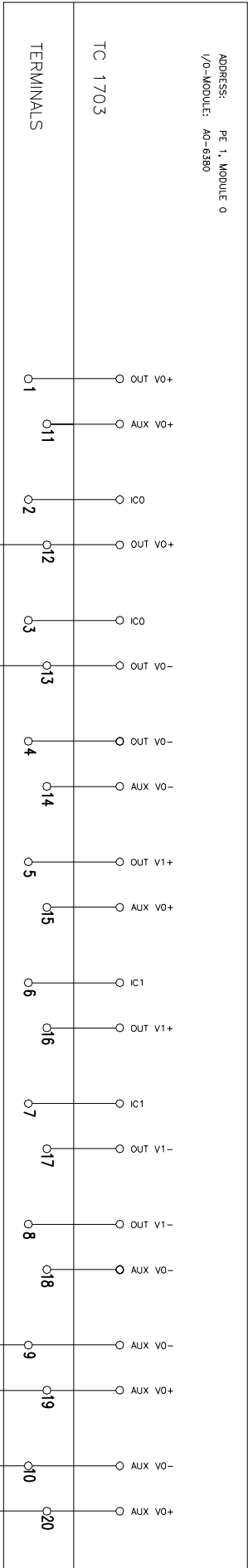
–A23

AO 0

0–10V
Control Signal Needle 2
0%.....100%
SIG_N2

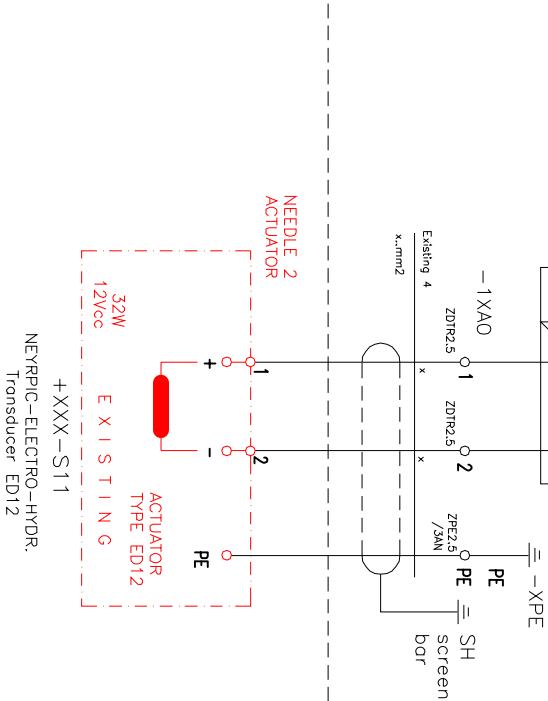
AO 1

0–10V
SPARE



INTERNAL

EXTERNAL



–A12

AO 2

0–10V

Control Signal Needle 3

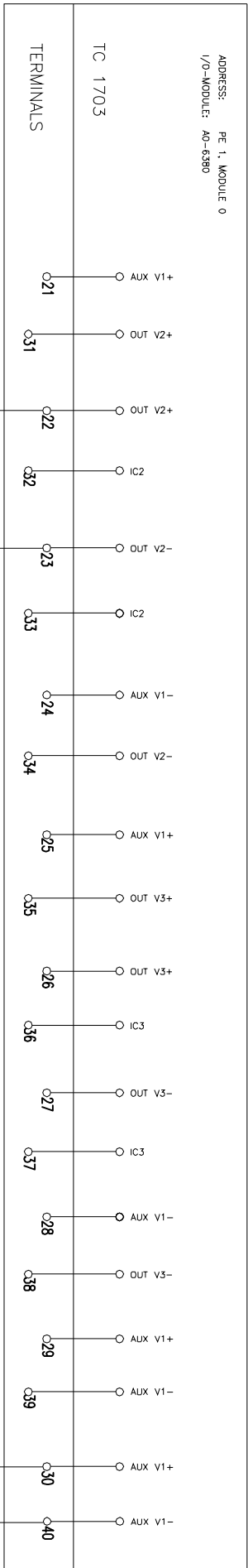
0%...+100%

SIG–N3

AO 3

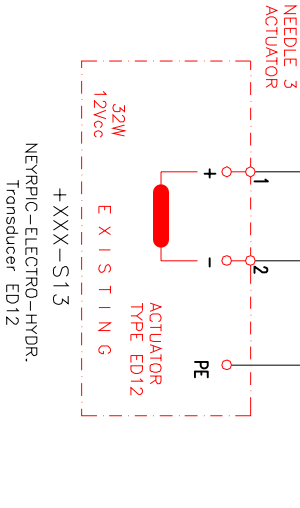
0–10V

S P A R E



INTERNAL

EXTERNAL



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				Ges./Appr. 20.06.2016 Fuchs		Projec – No :		Index		Bl./Sht. 44	
Änderung / Modification				Datum/Date		Bearb./Des.		Mossstab/Scale		von/of 69	
						Lost Edit: 20.07.2016		-----		A3	

ANDRITZ

HYDRO

Hydro

TURBINE CONTROL CUBICLE

Circuit Diagram

Roll #1

AO–6390

Module 0

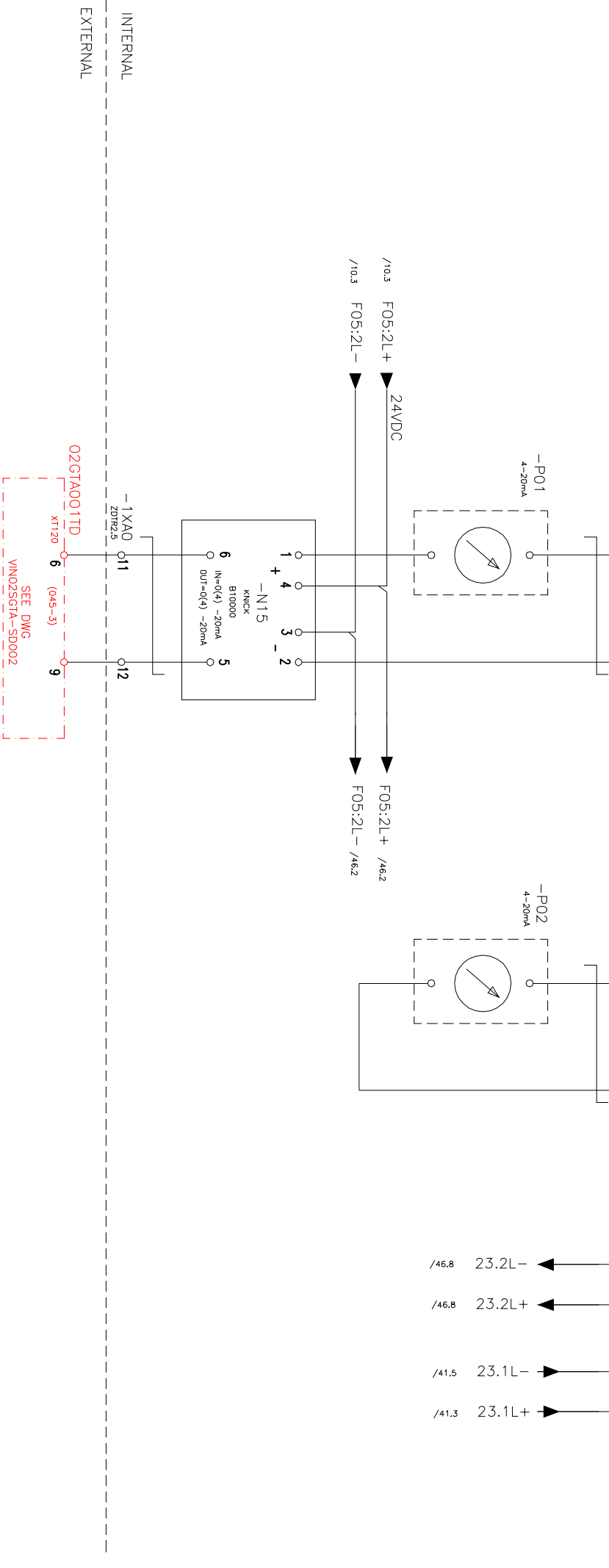
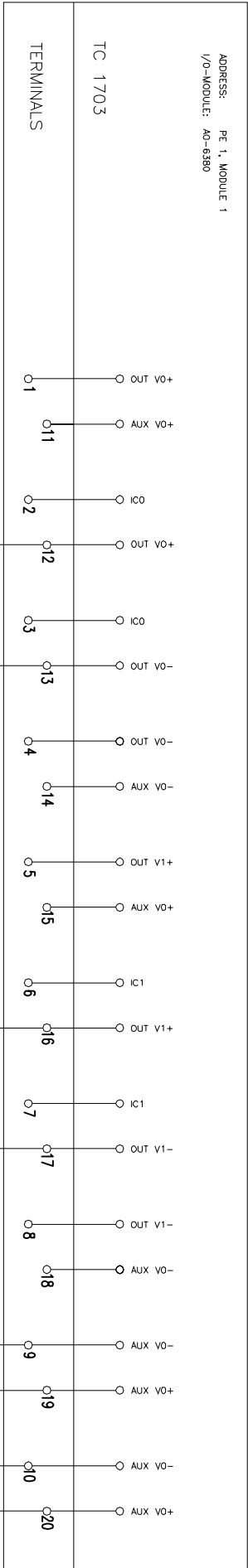
–A23

AO 0

4–20mA
Turbine Speed
0....200%
N_ACTIVE

AO 1

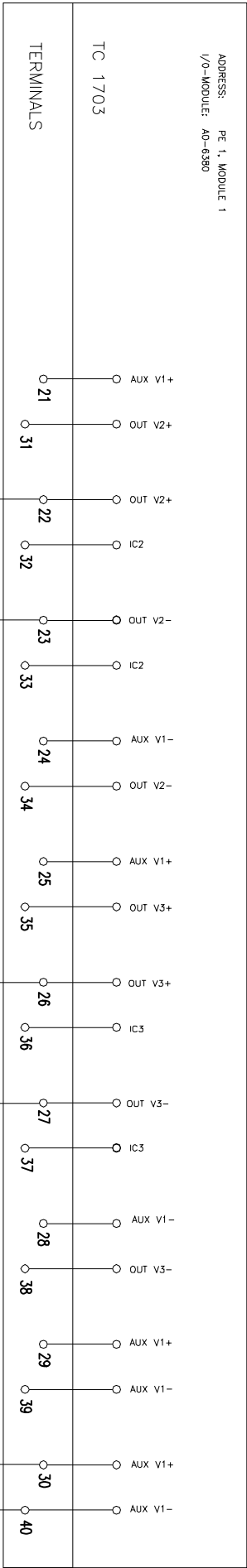
4–20mA
Load Setpoint



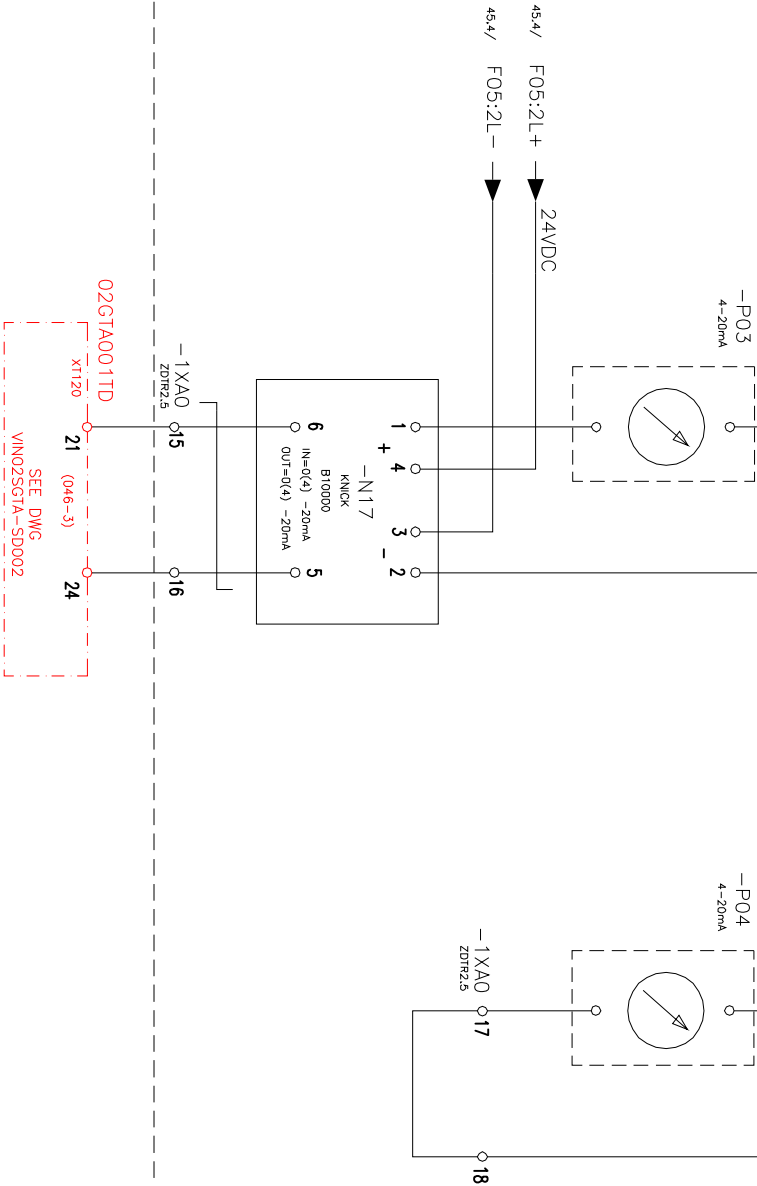
-A23

AO 2
4-20mA
Opening Limit
0...100%
OPL

AO 3
4-20mA
Speed Setpoint
-10...0...+10%



45.8/ 23.2L+ →
45.8/ 23.2L- →



0

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8

9

AI 0
4–20mA

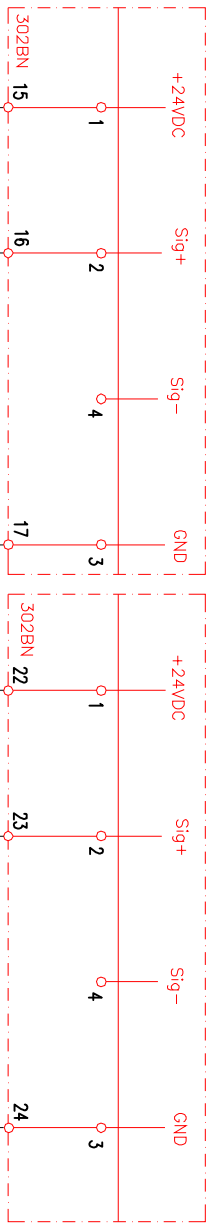
Feedback Signal Needle 2
0...100%
N2–SIGNAL

AI 1
4–20mA

Feedback Signal Needle 3
0...100%
N3–SIGNAL

02GTA002CR

02GTA002CR

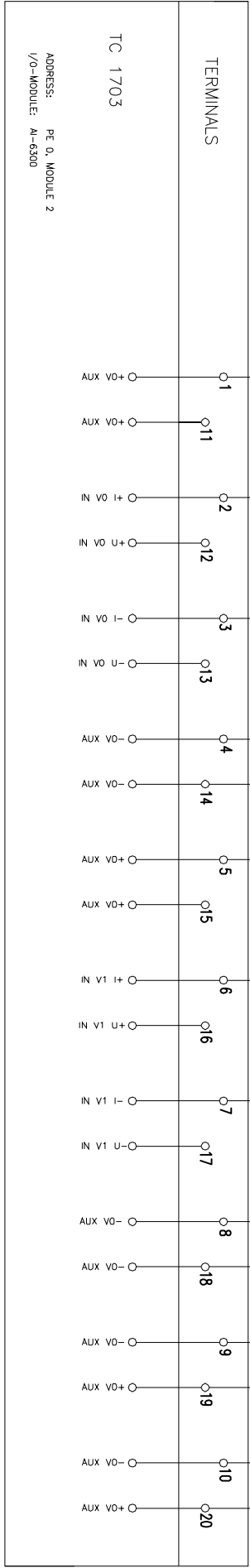


EXTERNAL
INTERNAL

INTERNAL

N28:8– /24V
N28:7+ /24V

–A24



ADDRESS: PE 0, MODULE 2
I/O-MODULE: AI-6300

TC 1703

AUX V0+

AUX V0+

IN V0 +

IN V0 +

IN V0 –

IN V0 U–

AUX V0–

AUX V0–

AUX V0+

AUX V0+

IN V1 +

IN V1 U+

IN V1 –

IN V1 U–

AUX V0–

AUX V0–

AUX V0–

AUX V0+

AUX V0–

AUX V0+

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Patentreferenz: 1141 Vienna Austria-rl, project, note ISO 16016
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Patentreferenz: 1141 Wien, ÖSTERREICH-Schutz, ISO 16016 beachten

Gez./Draw

Gez./Check

Gez./Appr.

Mossstab/Sole

Gez./Draw

Gez./Check

Gez./Appr.

Mossstab/Sole

Kunde/Client:

Anlage/Plant:

Projec – No :

Lost Edit: 11.05.2012

TURBINE CONTROL CUBICLE

Circuit Diagramm

Roll #1

AI-6300 Module 2

InL.No

= VINH SON

+MET100H100

Index

Bl./Sht. 47

von/of 69

A3

Zeich./Draw.No.

VNI2-COV-0021

Index

Bl./Sht. 47

von/of 69

A3

Zeich./Draw.No.

VNI2-COV-0021

Index

Bl./Sht. 47

von/of 69

A3

Zeich./Draw.No.

VNI2-COV-0021

Index

Bl./Sht. 47

von/of 69

A3

Zeich./Draw.No.

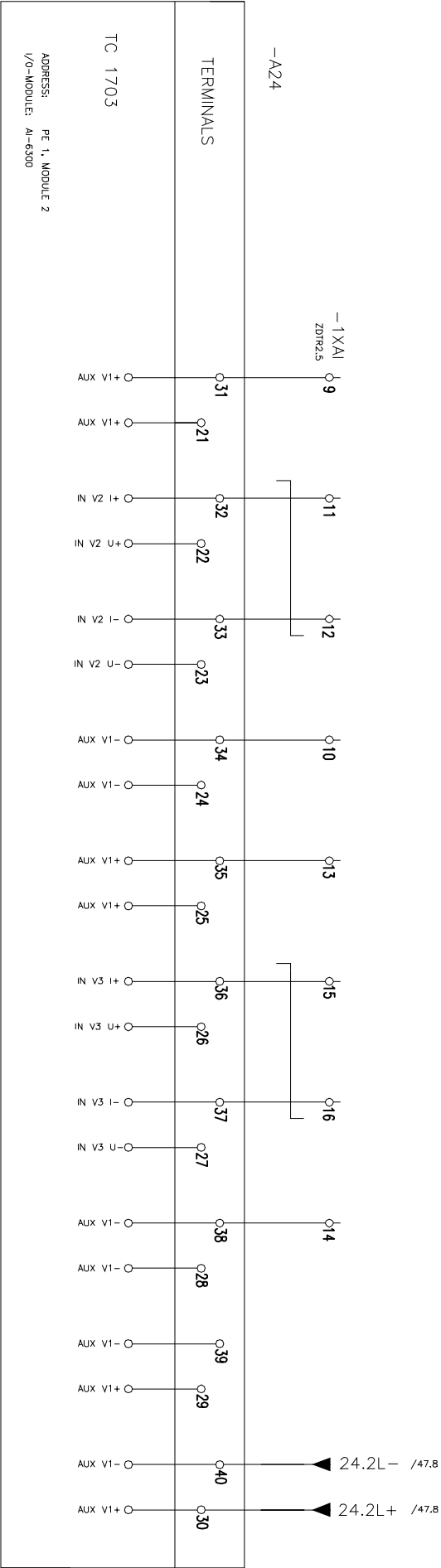
VNI2-COV-0021

AO 2
4 – 20mA
SPARE

AO 3
4 – 20mA
SPARE

EXTERNAL

INTERNAL



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				Gepr./Check	20.06.2016	Fuchs	Anlage/Plant: VINH SON							
				Ges./Appr.	20.06.2016	Fuchs	Projec - No :							
				Mossstab/Scale		-----	Lost Edit: 11.05.2012							
Änderung / Modification				Datum/Date	Beorb./Des.									
				ANDRITZ HYDRO										
				Turbine Control Cubicle										
				Circuit Diagram										
				Ref #1										
				AI-6300 Module 2										
				Int.No		= VINH SON								
				Zeich./Draw No.		+METROCH100								
				Index		Bl./Sht. 48								
						von/of 69								
				A3										

SMS 0

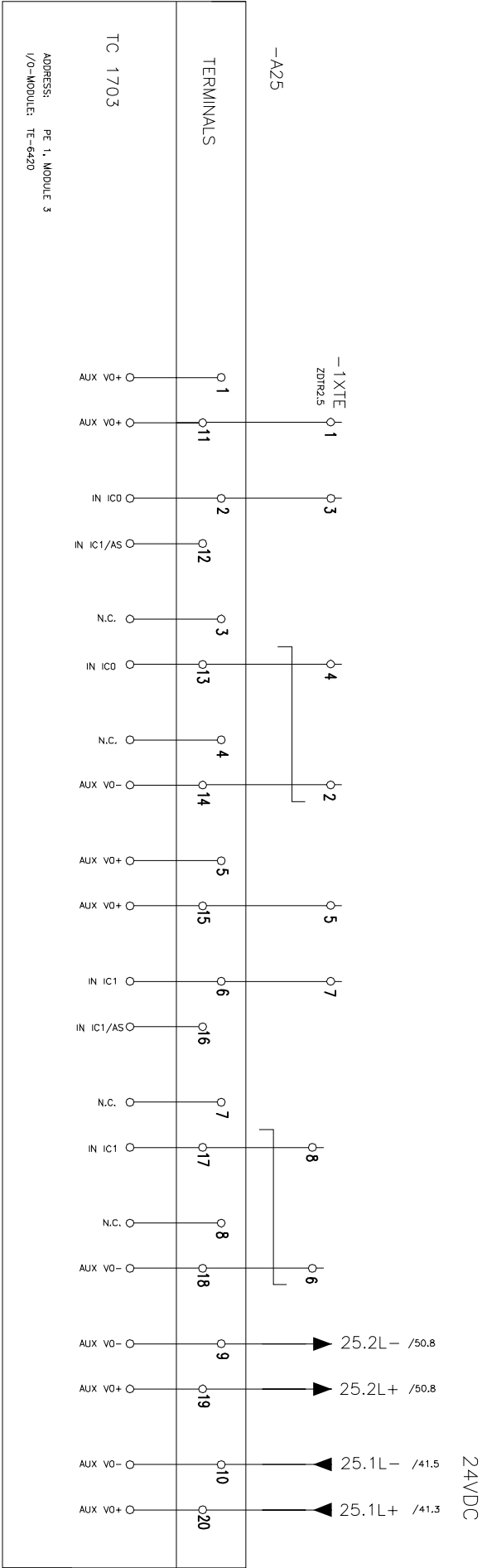
SPARE

SMS 1

SPARE

EXTERNAL

INTERNAL

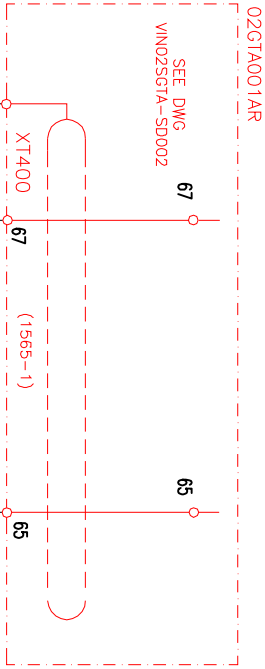
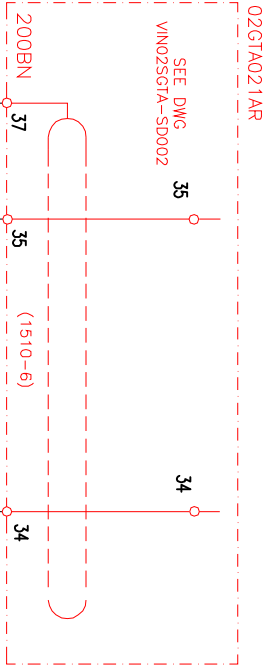


SMS 2

Generator Frequency
0...100Hz = 0...200%
n2_SMS

EXTERNAL

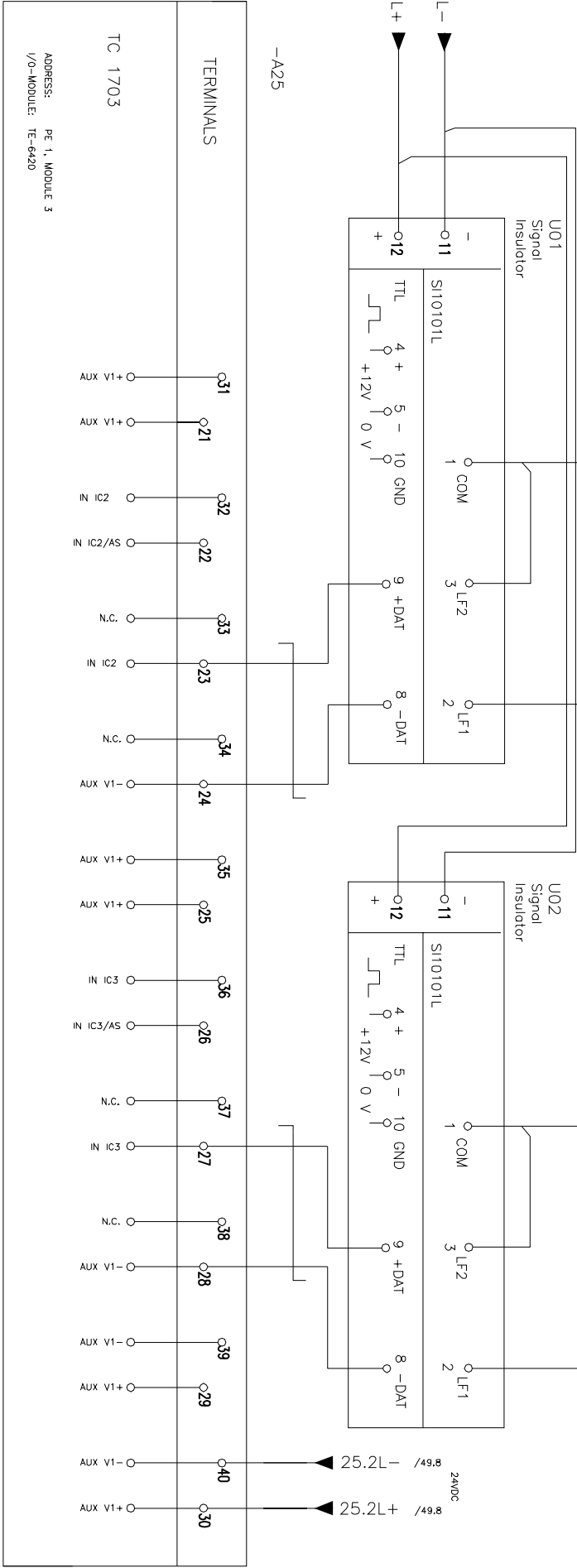
INTERNAL



SMS 3

Line Frequency
0...100Hz = 0...200%
n4_SMS

-A25



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Perzogenitz/26 1141 Vienna Austria-rc, project, note ISO 16016				20.06.2016				Auer			
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Perzogenitz/26 1141 Wien, ÖSTERREICH-Schultr. ISO 16016 beachten				20.06.2016				Fuchs			
				Ges./Appr.				Projec - No :			
				20.06.2016				Fuchs			
Änderung / Modification				Datum/Date				Mossstab/Scale			

				Beerb./Des.				Lost Edit: 20.07.2016			

ANDRITZ

Hydro

HYDROTECHNI

TURBINE CONTROL CUBICLE				InL.No				= VINH SON			
Circuit Diagramm				Zechn./Draw.No.				+METROCH100			
Reil #1				Index				Bl./Sht. 50			
TE-6420 Module 3				von/of 69				A3			

0

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DI 0 DI 1 DI 2 DI 3 DI 4 DI 5 DI 6 DI 7
Turbine SPARE SPARE SPARE SPARE SPARE SPARE
Controller
Stop

02GTA003AR

SEE DWG
VIN02SGTA-SD001

OXD:65 /32.8

5 782XR
3 (1108-B)

-S.
NO
C

-S.
NO
C

-S.
NO
C

-S.
NO
C

XT6110

165

EXTERNAL

INTERNAL

24VDC

-1 XDI
ZDTR2.5

-A26

TERMINALS

TC 1703

ADDRESS: PE 1, MODULE 4
I/O-MODULE: DI-6100

IN D00

AUX V0+ (-)

IN D01

AUX V0+ (-)

IN D02

AUX V0+ (-)

IN D03

AUX V0+ (-)

IN D04

AUX V0+ (-)

AUX V0+ (-)

IN D06

AUX V0+ (-)

IN D07

AUX V0+ (-)

AUX V0- (+)

AUX V0+ (-)

AUX V0- (+)

AUX V0+ (-)

AUX V0+ (-)

26.2L- /52.8

26.2L+ /52.8

26.1L- /41.5

26.1L+ /41.3

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Permitierung/Ze 1141 Vienna Austria-ru, project, note ISO 16016

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Permitierung/Ze 1141 Wien, ÖSTERREICH-Schulstr. 150 16016 besetzen

Gez./Draw

Auer

Gepr./Check

20.06.2016

Kunde/Client:

VINH SON

TURBINE CONTROL CUBICLE

Int.No

= VINH SON

+METROCH100

A3

Gepr./Check

20.06.2016

Fuchs

Mossstab/Scale

Lost Edit: 20.07.2016

Projec - No :

Projec

Zeich./Draw.No.

VNI2-COV-0021

Index

Bl./Sht. 51

von/of 69

A3

Änderung / Modification

Datum/Date

Bearb./Des.

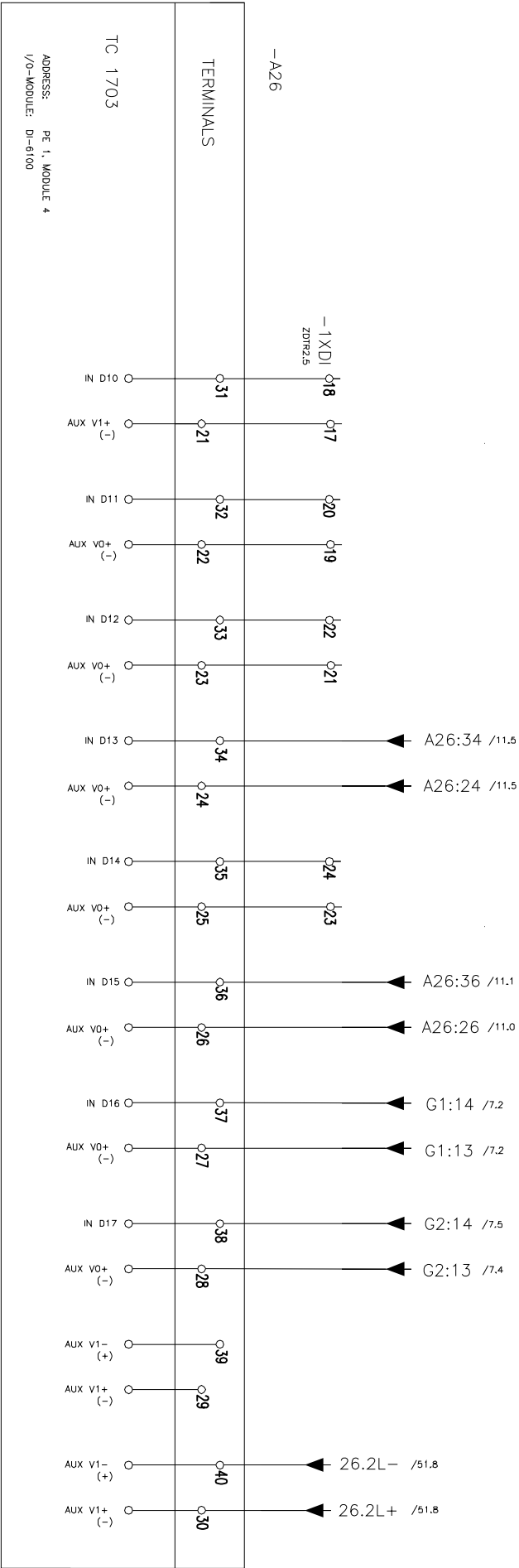
Mossstab/Scale

Lost Edit: 20.07.2016

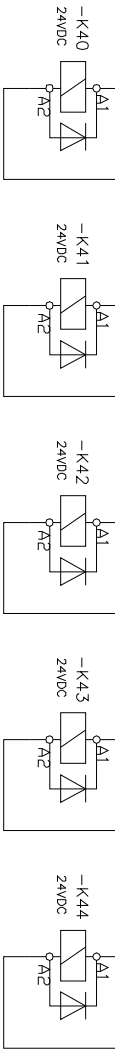
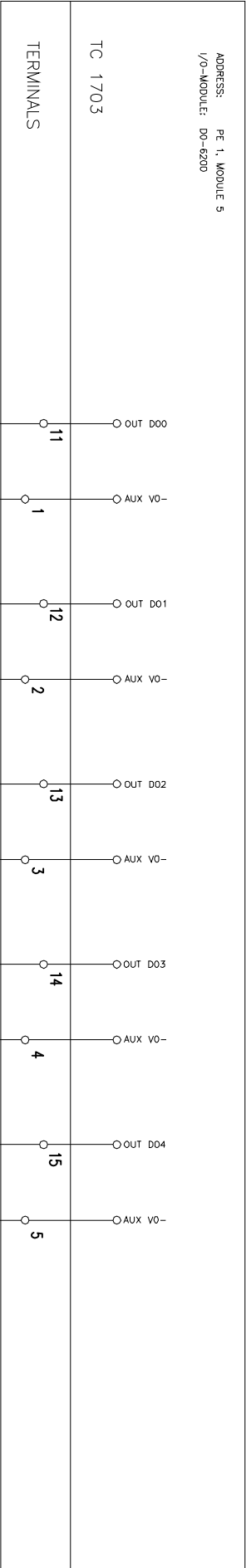
- | | | | | | | | |
|-------|-------|-------|-----------------------------|-------|----------------|----------------------------|----------------------------|
| DI 10 | DI 11 | DI 12 | DI 13 | DI 14 | DI 15 | DI 16 | DI 17 |
| SPARE | SPARE | SPARE | Load
Circuit
Controls | SPARE | MCB
Tripped | Power
Supply 1
Fault | Power
Supply 2
Fault |
| | | | F-FUSE | | F-FUSE | | F-PS2 |

EXTERNAL

INTERNAL

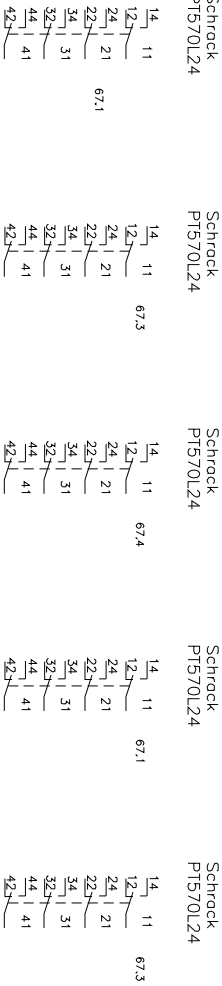


	DO 0	DO 1	DO 2	DO 3	DO 4
	PRIORITY	Needle 1	Needle 2	Needle 3	Needle 4
	NEEDLE 1	Positioner	Positioner	Positioner	Positioner
-A27	Faulty	Faulty	Faulty	Faulty	Faulty



INTERNAL

EXTERNAL



0

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9

D0 5

Deflector
Positioner
Faulty

D0 6

Circuit
Broker
Open

D0 7

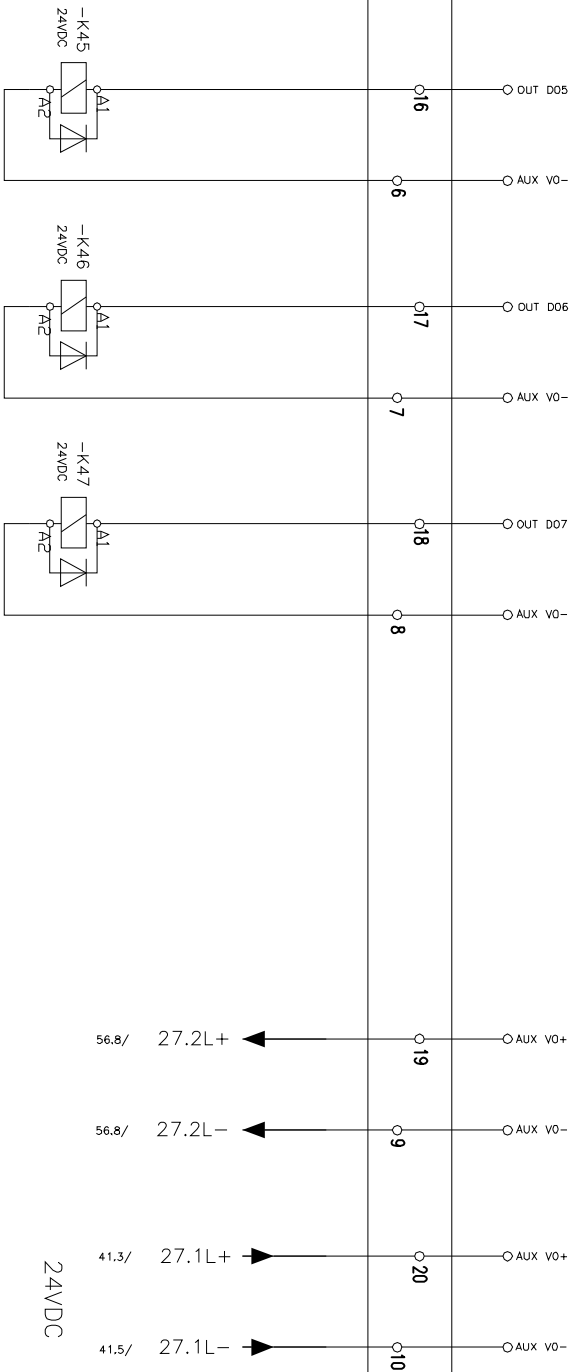
needle 1
closed

—A27

ADDRESS: PE 1, MODULE 5
I/O-MODULE: AO-6200

TC 1703

TERMINALS



INTERNAL

EXTERNAL

Schrack PT570L24	Schrack PT570L24	Schrack PT570L24
14 11 67.4	14 11 67.6	14 11 67.7
12 24 21	12 24 21	12 24 21
22 34 31	22 34 31	22 34 31
32 44 41	32 44 41	32 44 41
42 42 41	42 42 41	42 42 41

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ISO 16016 beachten										ANDRITZ Hydro										TURBINE CONTROL CUBICLE Circuit Diagramm Roll #1 DO-6200 Module 5										InL.No = VINH SON +METROH100										Zech./Draw.No. VN12-COV-0021										Index B1./Snt. 54 von/of 69										A3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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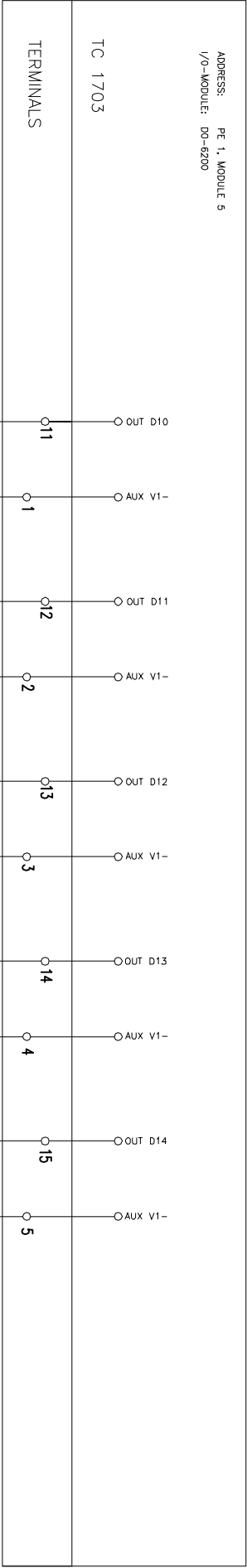


Deflector not blocked → DCS Gov Governor started

~~turbine started~~ turbine stopped

-A27

D0 10	D0 11	D0 12	D0 13	D0 14
needle 2	needle 3	needle 4		
closed	closed	closed		



INTERNAL

EXTERNAL

Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24
14 11 66.1	14 11 66.3	14 11 66.4	14 11 65.7	14 11 65.1
12 24 21	12 24 21	12 24 21	12 24 21	12 24 21
34 31	34 31	34 31	34 31	34 31
32 44 41	32 44 41	32 44 41	32 44 41	32 44 41
42	42	42	42	42

DO 15

DO 16

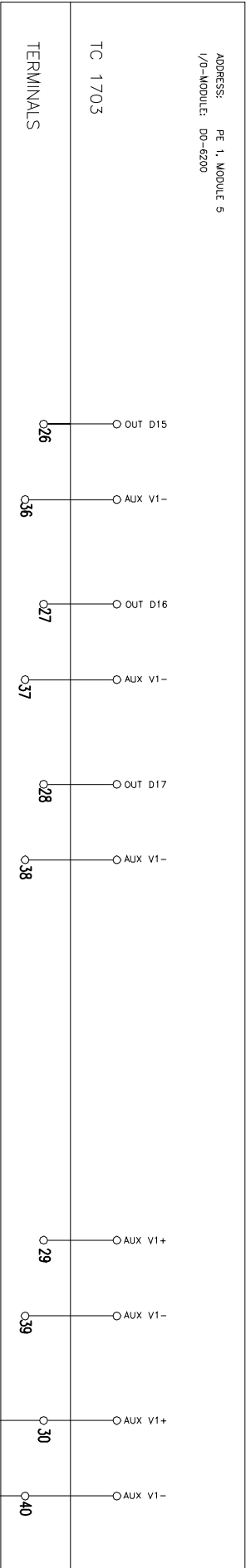
DO 17

SPARE

SPARE

SPARE

—A27



INTERNAL

EXTERNAL

0

1

2

3

4

5

6

7

8

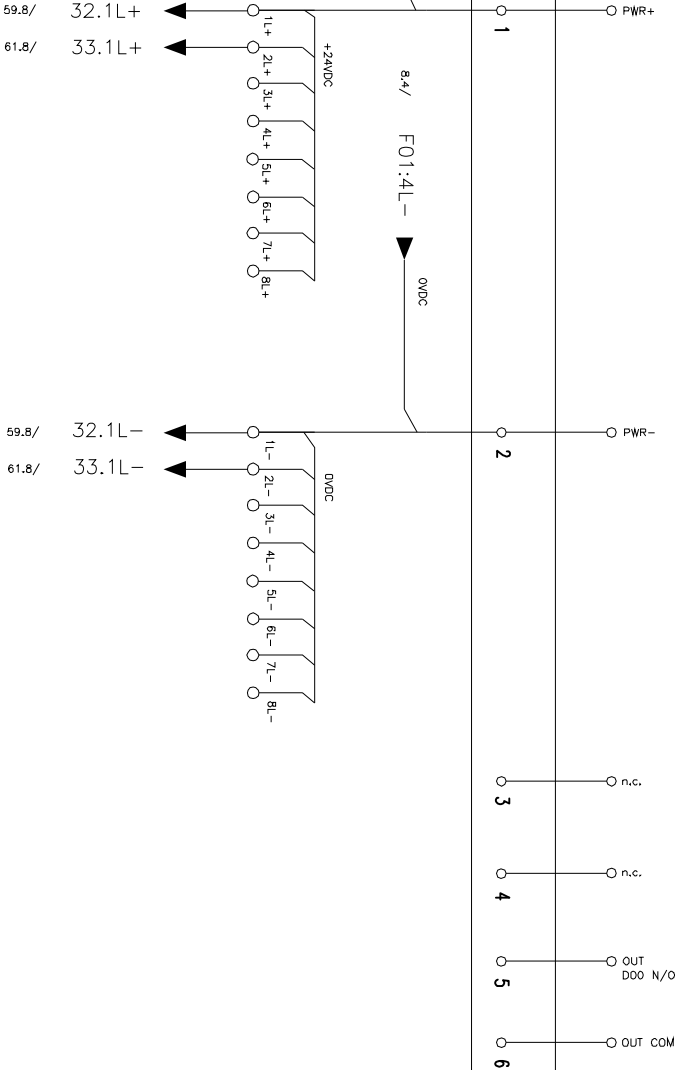
9

—A30

ADDRESS: PS 2, PS-MODULE
I/O-MODULE: PS-6630

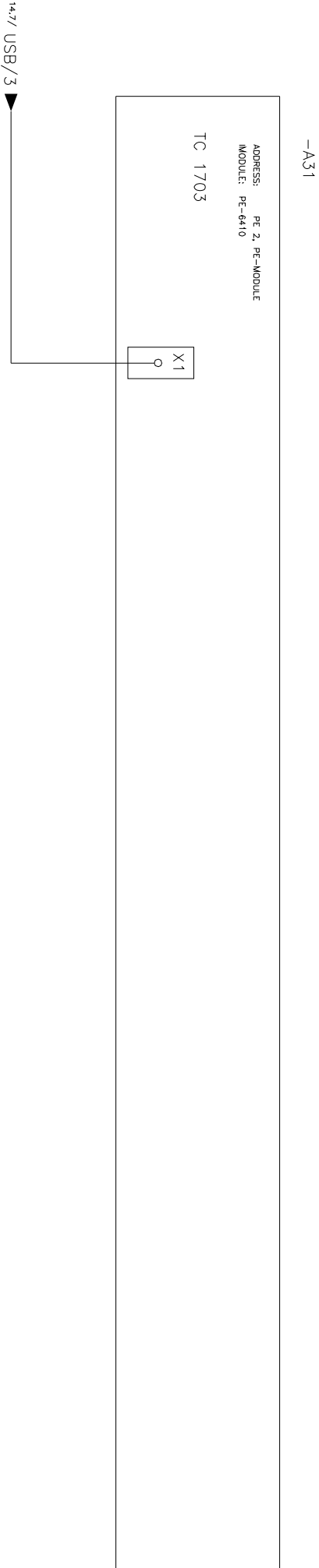
TC 1703

TERMINALS



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Gez./Draw		20.06.2016		Auer		Kunde/Client:													
Gepr./Check		20.06.2016		Fuchs		Anlage/Plant:		VINH SON											
Ges./Appr.		20.06.2016		Fuchs		Projec – No :													
Mossstab/Sole		-----				Lost Edit: 20.07.2016													
Änderung / Modification		Datum/Date		Beorb./Des.															





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				Gepr./Check	20.06.2016	Fuchs	Anlage/Plant:	VINH SON		
				Gepr./Appr.	20.06.2016	Fuchs	Projec - No :			
				Mossstab/Scale	-----		Lost Edit:	20.07.2016		
Änderung / Modification				Datum/Date	Bearb./Des.					

TURBINE CONTROL CUBICLE
Circuit Diagram
Peripherycontroller #3 – Rail 2

Int.No		= VINH SON	
Zeich./Draw.No.		+MET10GH100	
VN12-CON-0021		Index	Bl./Sht. 58
		von/of 69	
		A3	

–A32

AO 0

4–20mA

Position of Needle 1

0...100%

N1_INT

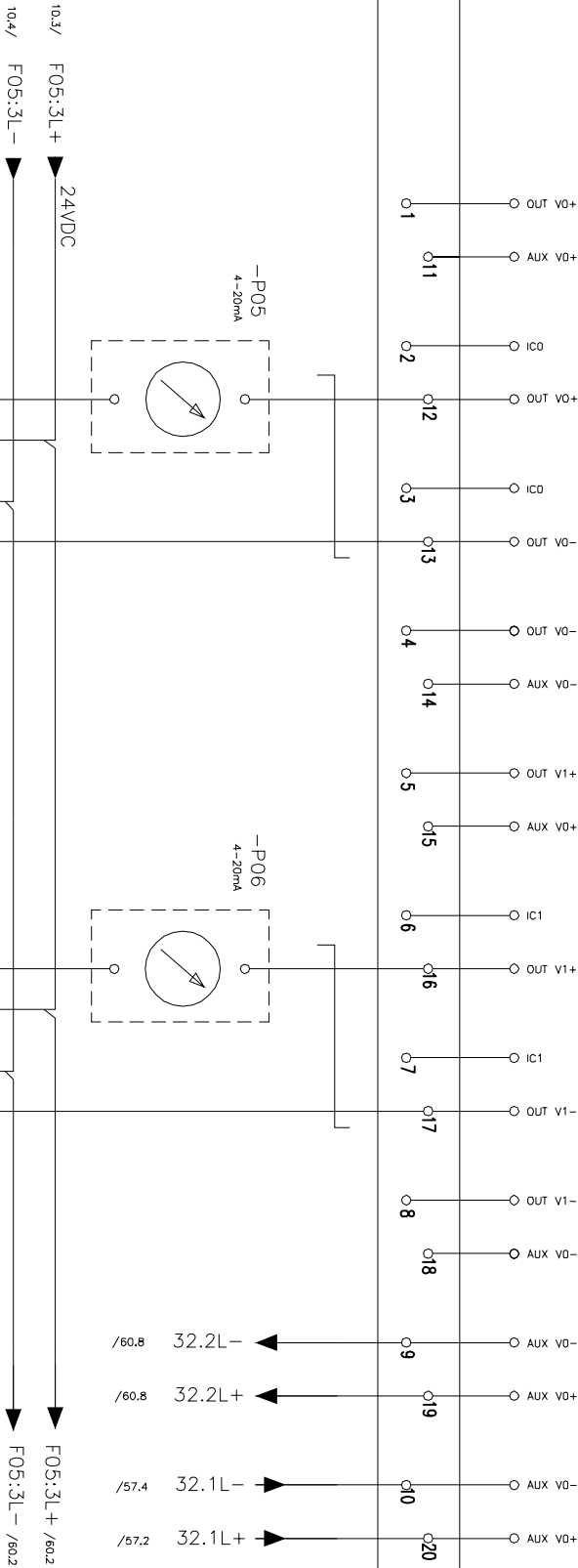
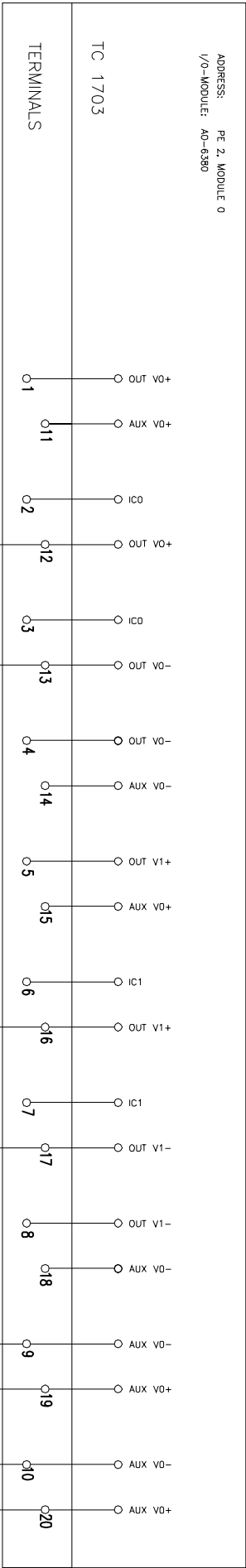
AO 1

4–20mA

Position of Needle 2

0...100%

N2_INT



INTERNAL

EXTERNAL

02GTA001TD
XT120 26 (046–5) 27
SEE DWG
VINO2SGTA–SD002

02GTA003AR
TBA1.03 7 (1113A–3) 8
SEE DWG
VINO2SGTA–SD002

AO 2

4-20mA

Position of Needle 3

0...100%
NIT INT

N3_INT

-A32

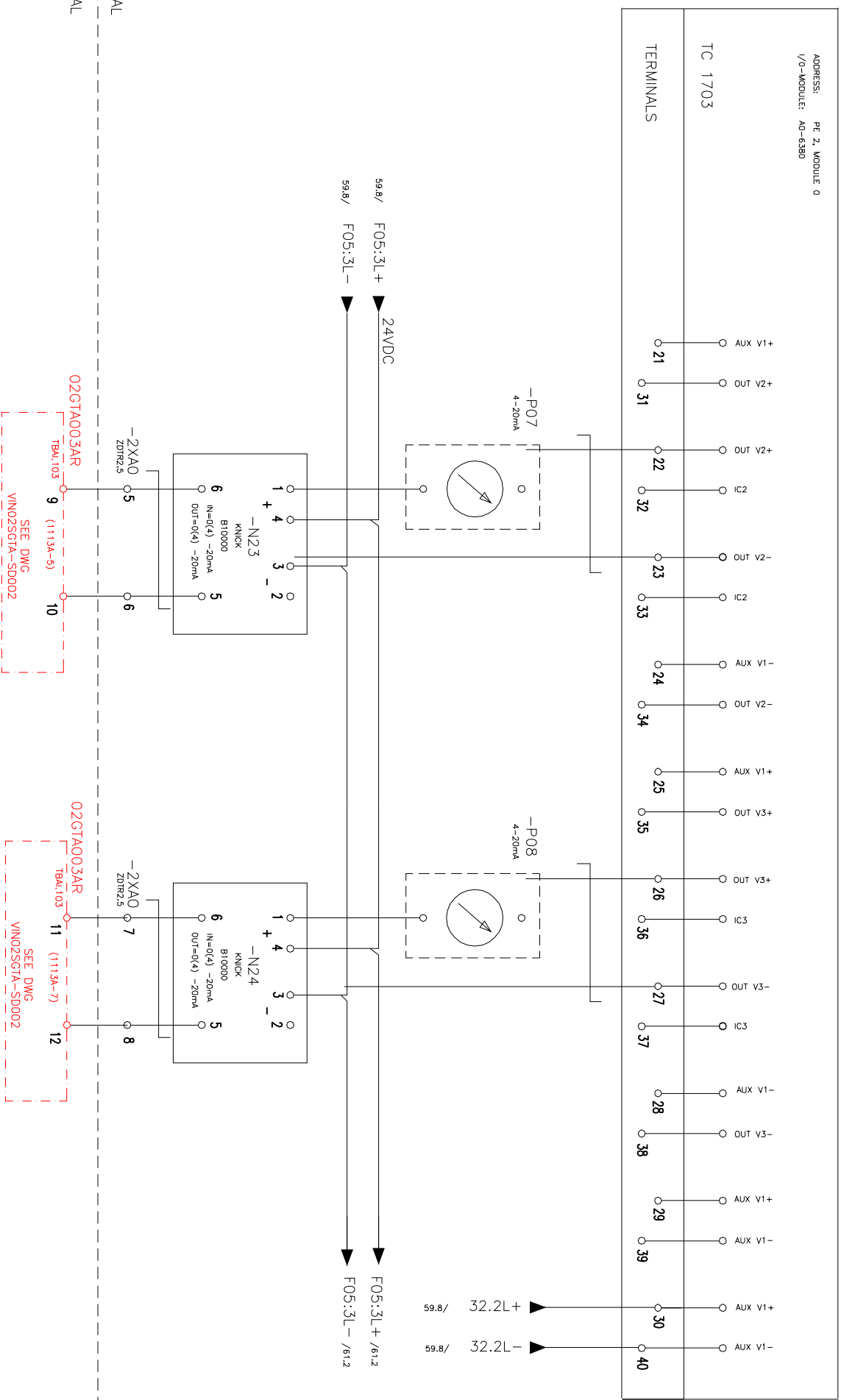
AO 3

4-20mA

Position of Needle 4

0..100%

N4_INT



ADDRESS: PE 2, MODULE 0
I/O-MODULE: AO-6380

TC 1703

TERMINALS

Figure 1 is a schematic diagram of a 40-channel system. It features a central vertical line with 40 horizontal branches, each ending in a circle containing a number from 1 to 40. The branches are labeled on the right side with various system components: AUX V1+, OUT V2+, OUT V2+, IC2, OUT V2-, IC2, AUX V1-, OUT V2-, AUX V1+, OUT V3+, OUT V3+, IC3, OUT V3-, IC3, AUX V1-, OUT V3-, AUX V1+, AUX V1-, AUX V1+, and AUX V1-.

59.8/ 32.2L+
59.8/ 32.2L-

24VDC

598/ F05:3L+ F05:3L+ / 61.2

598/ F05:3L- F05:3L- / 61.2

INTERNAL
EXTERNAL

02GTA003AR

TBA.103 9 (1113A-5) 10

SEE DWG
VIN02GTA-SD002

02GTA003AR

TBA.103

11 (1113A-7)

12

SEE DWG
VIN02SGTA-SD002

0

1

2

3

4

5

6

7

8

9

–A33

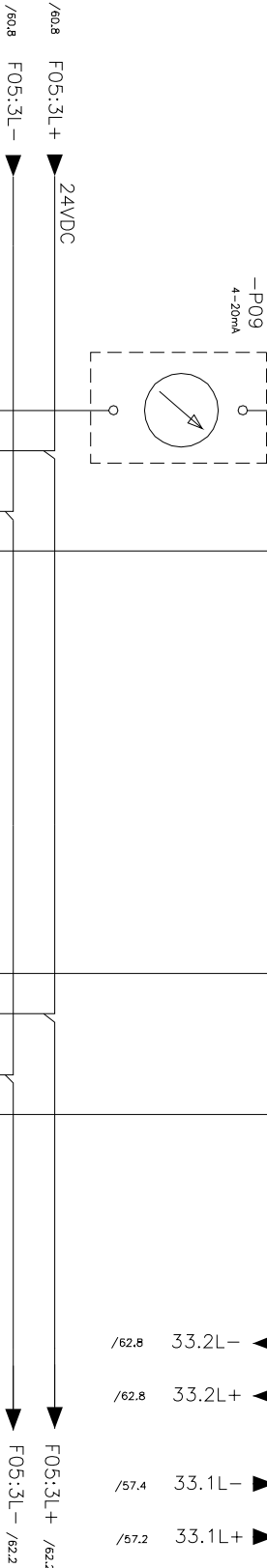
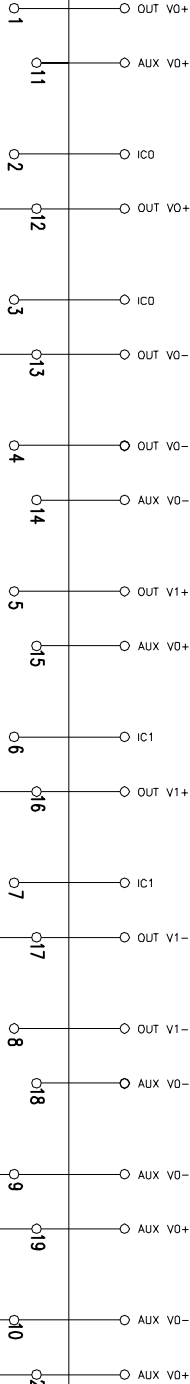
AO 0
4–20mA
Actual Power
0...50MW
P_ACTIVE

AO 1
4–20mA
Frequency/Load
indication

ADDRESS: PE 2, MODULE 1
I/O-MODULE: AO-6380

TC 1703

TERMINALS



INTERNAL

EXTERNAL

02GTA0011TD
XT120 11 (045-5) 12
SEE DWG
VIN02SGTA-SD002

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				Geop./Check	20.06.2016	Fuchs	Anlage/Plant:					Circuit Diagramm	Zech./Draw No.	+MET008100	
				Ges./Appr.	20.06.2016	Fuchs	Projec - No :					RoI #2	Index	Bl./Sht. 61	
				Mossstab/Scale	-----		Lost Edit: 11.05.2012					AO-6380	Module 1	von/of 69	
Änderung / Modification				Datum/Date	Beorb./Des.									A3	

AO 2

4-20mA

Position of Deflectors

0%...+100%

DE-INT

AO 3

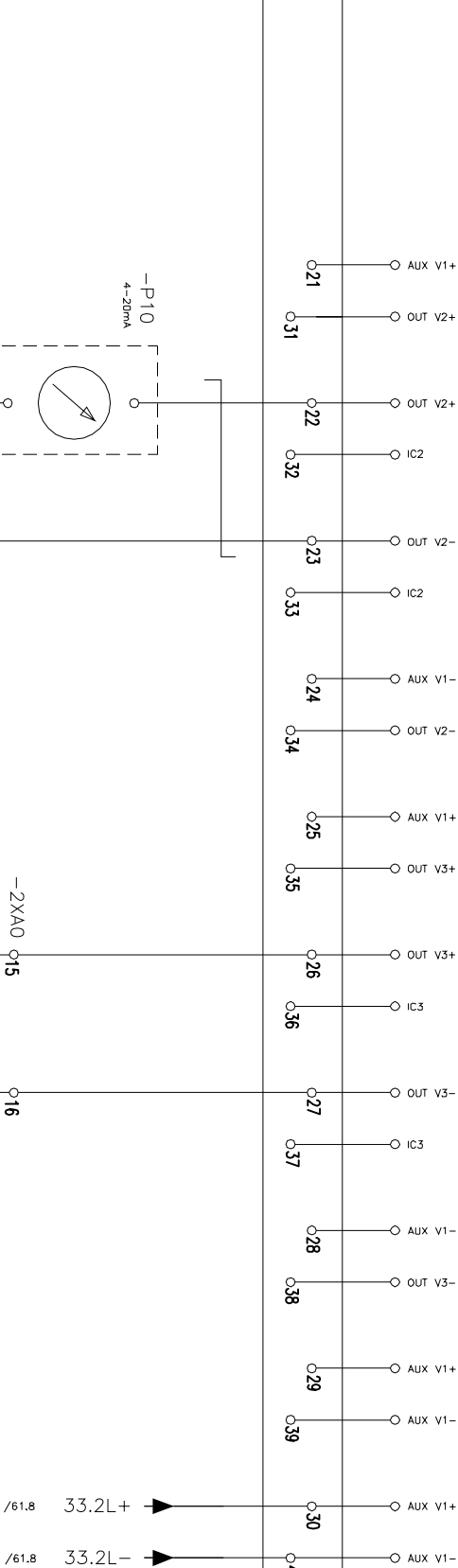
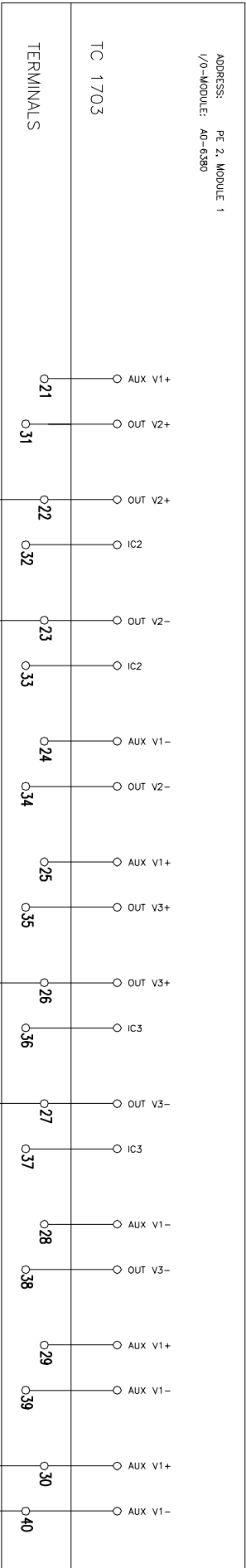
4-20mA

Regulating Device

Indicator

-A33

ADDRESS: PE 2, MODULE 1
I/O-MODULE: AO-6380



INTERNAL

EXTERNAL

02GTA001TD

X1120 16 (045-7) 17

SEE DWG
VIN02SG1A-SD002

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				Gepr./Check	20.06.2016	Fuchs	Anlage/Plant:	VINH SON			Zech./Draw No.		+MET008100	
				Ges./Appr.	20.06.2016	Fuchs	Projec - No :				Index		B1./Sht. 62	
				Mossstab/Sole	-----		Lost Edit: 11.05.2012				VIN12-CON-0021		von/of 69	
Änderung / Modification				Datum/Date		Beorb./Des.							A3	

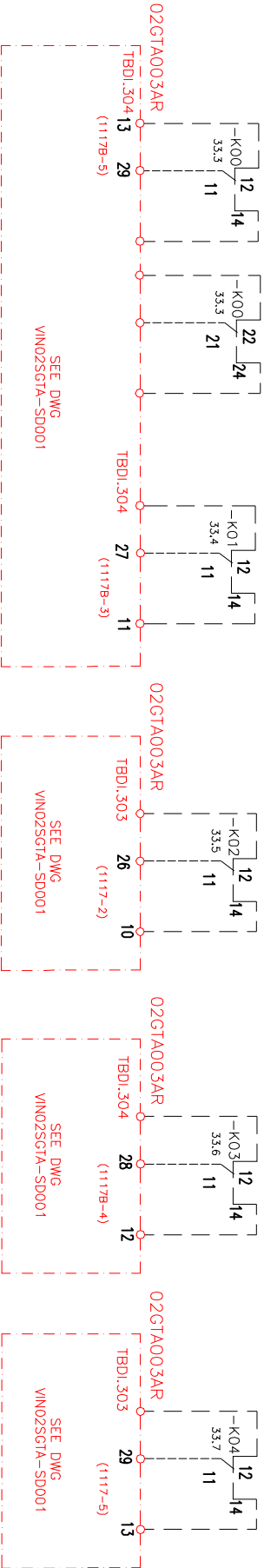
Major Fault
No Turbine Governor
Shut Down

Minor Fault
General
Alarm

Turbine
Limited

Isolating
Operation
Active

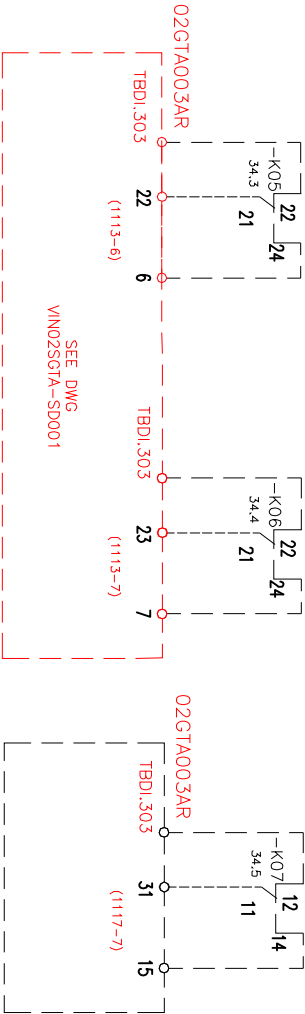
Speed
Controller
Active



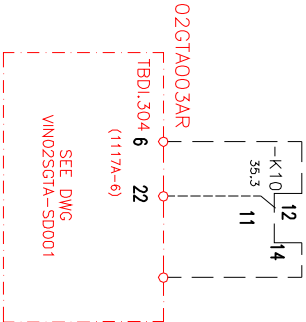
PRIORITY NEEDLE 2

PRIORITY NEEDLE 3

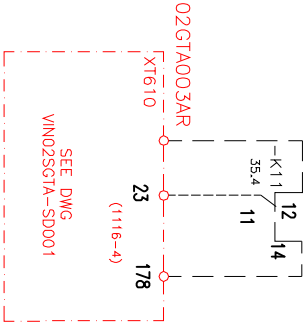
TC Ready
To Start



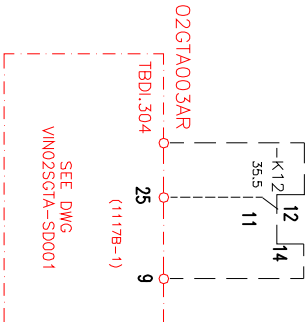
Speed Switch
n > 80%



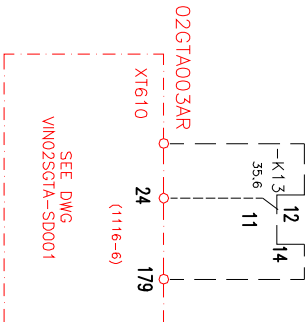
Speed Switch
Brake
n > 10%



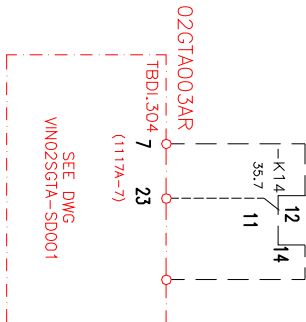
Speed Switch
Stonstill
n > 1%



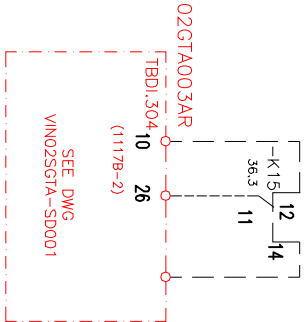
Speed Switch
Brake
n > 15%



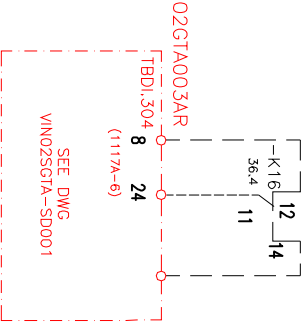
Speed Switch
Excitation
n > 70%



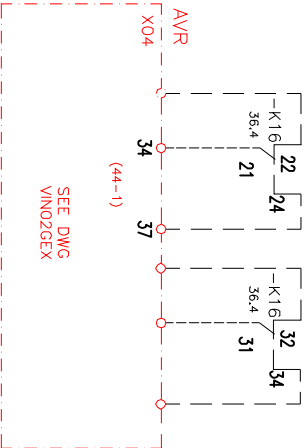
Speed Switch
Overspeed
n > 120%



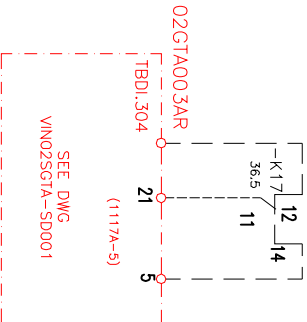
Speed Switch
n > 90%



SPARE



Speed Switch
SPARE



turbine
stoped

spare

Needle 3
not bloked

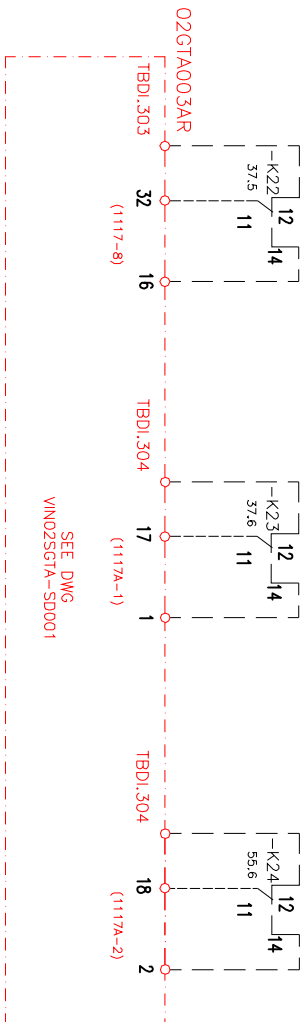
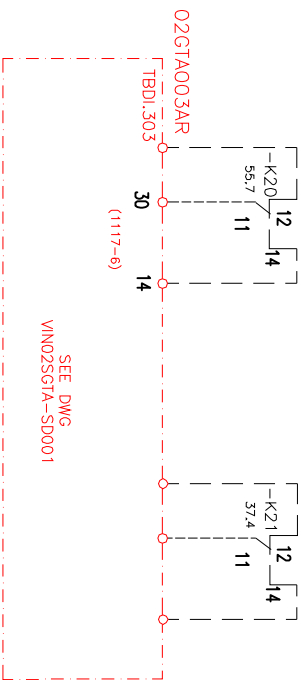
Needle 4
not blocked

~~turbine
started~~

Governor started

Deflector
not bloked

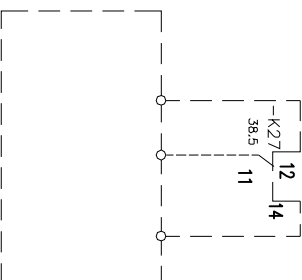
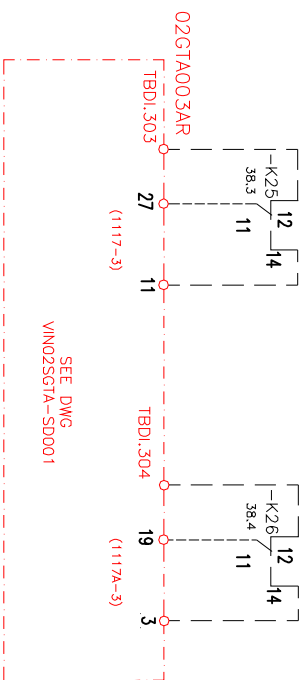
(Hàng tin DES)



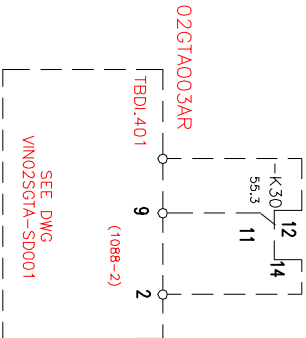
Opening Limited
at Max. Position

Power Less
Than 1MW

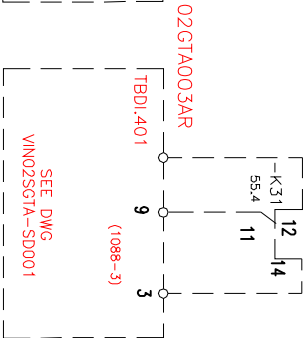
Governor in
Automatic Mode
spare



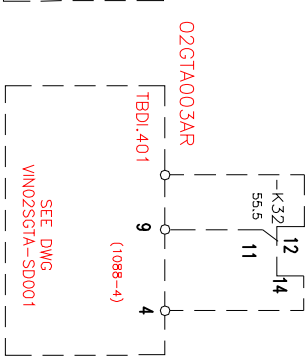
needle 2
closed



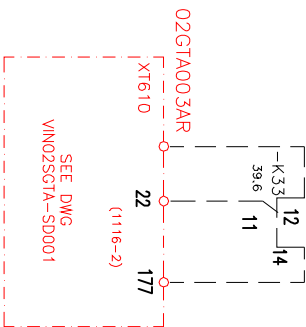
needle 3
closed



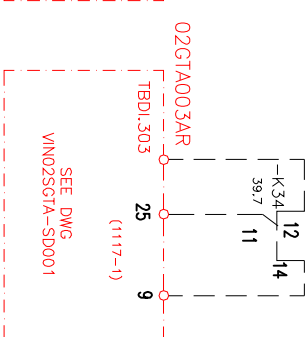
needle 4
closed



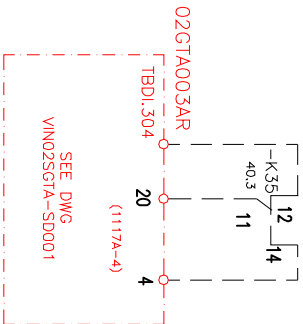
speed
signal
faulty



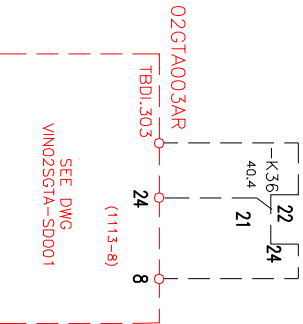
Speed NoLoad
position



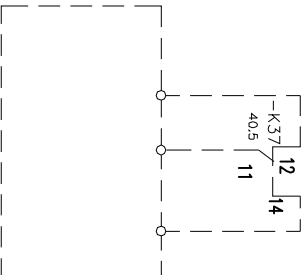
Start LO_ON
feedback



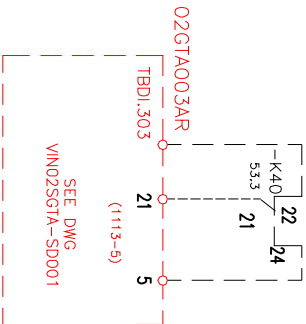
PRIORITY NEEDLE 4



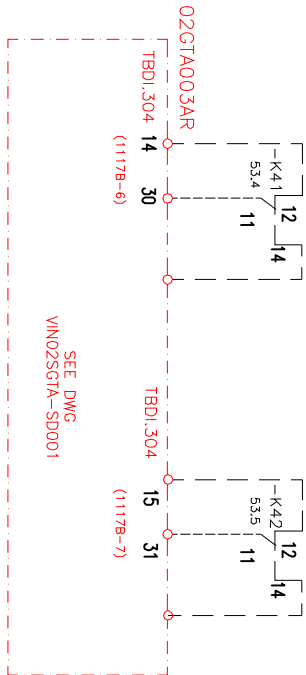
Nozzles
Are Closed
spare



priority
needle 1

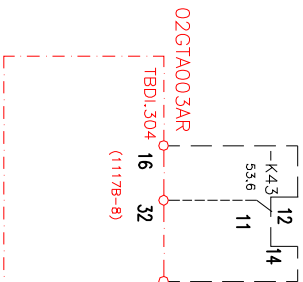


Needle 1
Positioner
Faulty

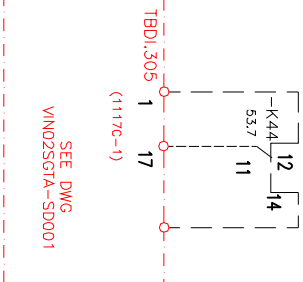


Needle 2
Positioner
Faulty

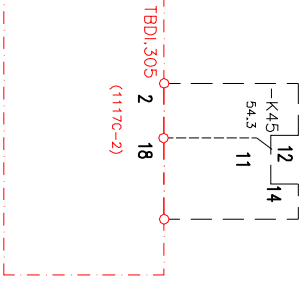
Needle 3
Positioner
Faulty



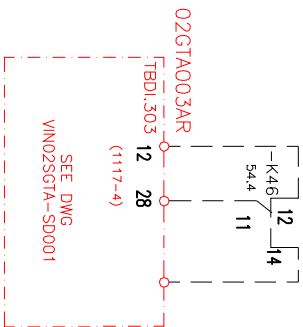
Needle 4
Positioner
Faulty



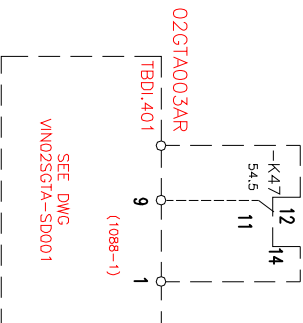
Deflector
Positioner
Faulty



Circuit
Broker
Open



needle 1
closed



[illegible]